

# RFQ: Stack Trading Prop Tech V7

Updates in V7:

1. Correctly naming AWS SES instead of Sendgrid
2. Added BPS endpoint to support changing Futures platform assigned.
3. Supporting the choice of 3 packages in the checkout
4. Adding 3.8 MT5 Plugin/Server-side code required

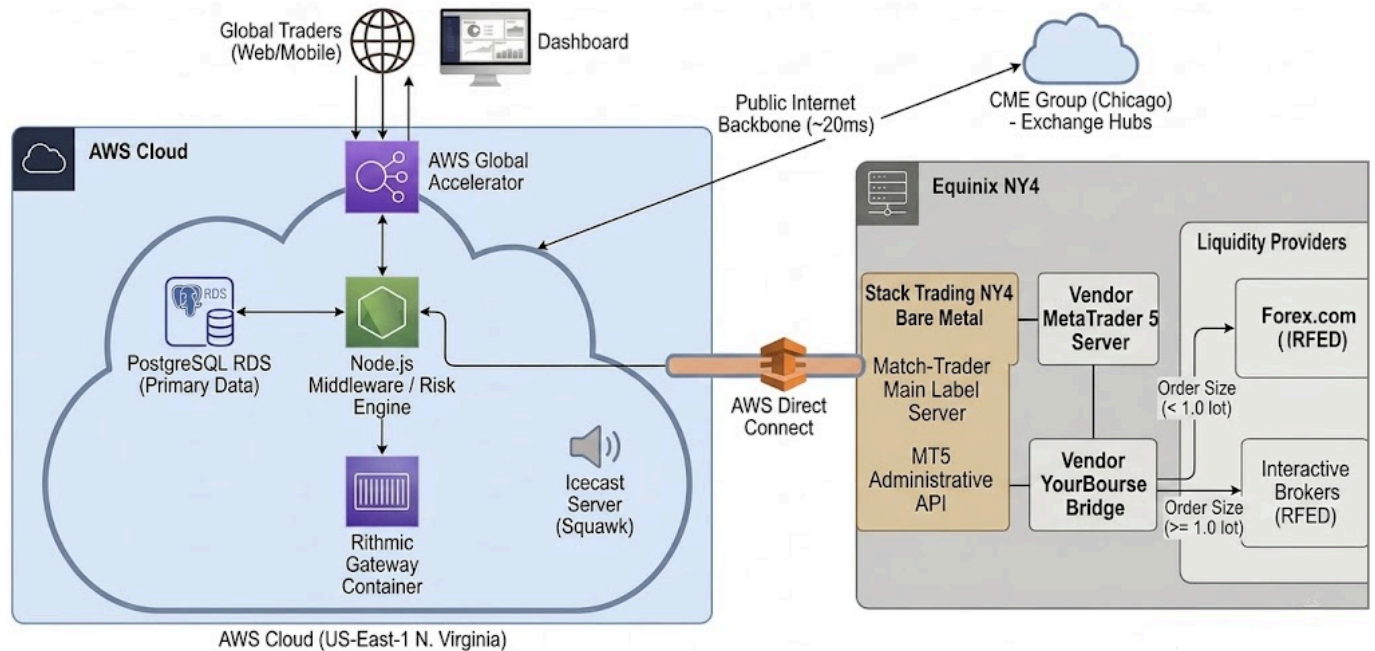
## 1.0 Executive Summary

Stack Trading is building a Direct Market Access (DMA) ecosystem designed to identify and capitalize professional trading talent across Futures and Forex. . The system operates on a "Simulation-First" model, monitoring trader performance in a demo environment and algorithmically promoting qualified users to a live, regulated trading environment (Stack Trading Brokerage LLC).

The target production deployment is a **hybrid topology**: (1) the administrative state, dashboard/API, and cloud-native orchestration run in **AWS**, and (2) the latency-sensitive **Forex execution core** runs on dedicated infrastructure in **Equinix NY4 (New York)** with physical cross-connects to execution vendors / liquidity providers and private connectivity back to AWS. The solution must be engineered for production-grade security controls, operational observability, and regulated audit/record retention.

The architecture also utilizes a Decoupled philosophy , separating business logic (Administrative State) from risk logic (Execution Gateways). The selected vendor will act as the **Systems Integrator**. Responsibilities include constructing the middleware layer, orchestrating state transitions via the Zapier automation bus, and integrating Rithmic, MetaTrader 5 (MT5) Manager API, TraderEvolution Admin APIs, and the YourBourse Bridge environment (FIX routing and trade translation). The integration with CQG and TT are **OUT OF SCOPE**.

In addition, the vendor must deliver a production-ready **security + availability architecture** (edge/WAF/DDoS strategy, IAM least privilege, encryption, audit logging, HA/DR approach, monitoring/alerting) and present **cost-aware scaling options** for AWS and the edge layer.



NOTE: StoneX is negotiating commissions to consolidate the Forex Flow so that the YourBourse Bridge may not have to route differently depending on lot size.

## 2.0 Technical Architecture Specification

### 2.1 Infrastructure & Hosting (AWS)

The system utilizes a hybrid architecture to optimize for the specific latency and OS requirements of the execution platforms. AWS hosts the scalable administrative control-plane (API, workflows, data, dashboards), and Equinix NY4 hosts the Windows-based Forex execution footprint. The baseline deployment stack includes edge security, traffic steering, an AWS load balancer/ingress layer, container orchestration, managed database/storage, secrets management, and private connectivity to NY4. Vendors may propose equivalent AWS-native alternatives that reduce cost/complexity while meeting performance, security, and reliability requirements.

| Component                    | Location & Type | Purpose & Connectivity                                                                                                                              |
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| Frontend CDN & Edge Security | Cloudflare      | Caches the static dashboard assets globally. Provides Layer 7 Web Application Firewall (WAF) and DDoS mitigation for all public-facing web domains. |

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| Network Acceleration | AWS Global Accelerator      | Optional traffic steering layer providing static anycast IPs and regional failover for the API ingress. If Cloudflare is deployed as the reverse proxy/WAF for the API domain, Global Accelerator may be used as the <b>origin target</b> (Cloudflare → Global Accelerator → ALB). If Cloudflare is <b>not</b> proxying the API domain, the vendor must implement equivalent WAF/DDoS and geo controls via AWS-native services (AWS WAF/Shield/CloudFront) and document the chosen design.                                                                                                          |
| Core Compute         | AWS US-East-1 (N. Virginia) | <p><b>AWS container orchestration for the middleware, Discord bot, workers, and gateway wrappers. Must support horizontal auto-scaling, rolling/blue-green deployments, and multi-AZ operation. Preferred: EKS with Fargate profiles; acceptable alternative: ECS/Fargate if the vendor can reduce operational overhead while meeting requirements.</b></p> <p><b>AWS EKS (Fargate), EKS or ECS/Fargate. Hosts the Node.js Middleware, Redis Cache, and Futures Gateway</b> (Rithmic wrappers).</p> <p><b>Connectivity:</b> Connects to Chicago Exchange Hubs via AWS Public Internet Backbone.</p> |

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| Forex Server         | Equinix NY4 (New York)       | <p><b>Dedicated Windows Server.</b> Hosts the TraderEvolution Main Label Server, the MT5 Manager API Gateway, and the YourBourse Bridge Connector for zero latency execution.</p> <p><b>Connectivity:</b> Physical Cross Connect to the YourBourse Bridge Server (NY4). Connects to Core Compute (AWS) via AWS Direct Connect private fiber. Inbound internet traffic is secured via Cloudflare Argo Tunnel (Outbound only connection).</p> <p><b>Vendor Servers (NY4).</b> The YourBourse Bridge and the MT5 Server equipped with the YourBourse Contest Plugin are hosted on separate vendor servers.</p> <p><b>Connectivity:</b> Direct connect between the MT5 Server and the YourBourse Bridge Server. Physical Cross Connect from the YourBourse Bridge Server to Forex.com and Interactive Brokers.</p> <p><i>NOTE: We are still negotiating for Forex clearing and ideally avoiding the need to bifurcate.</i></p> <p><b>Execution Accounts:</b> Forex.com and Interactive Brokers process executions exclusively through a Stack Trading Omnibus corporate account.</p> |
| Operational Database | AWS US-East-1 (CN. Virginia) | <p><b>Amazon RDS (PostgreSQL).</b> Multi-AZ deployment. The Single Source of Truth for all user and financial state. Data processing logic occurs here to minimize load on the automation bus.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Storage & Audit      | Amazon S3 (Glacier)          | <p>Immutable Storage. configured with Object Lock. Stores all raw trade logs, audit trails, and daily P&amp;L snapshots for compliance/regulatory retention. Stores</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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|                        |                            | trading plan image uploads and chart markups submitted via Typeform.                                                                                                                            |
| <b>Squawk Relay</b>    | <b>AWS EKS POD (Linux)</b> | Audio Stream Server. Hosts an Icecast 2 server. Ingests the authenticated Newsquawk URL and relays it to frontend listeners via low-latency HTTP streaming. (Prioritizes speed over buffering). |
| <b>Version Control</b> | <b>Github</b>              | Code storage with Github Actions to help manage CI/CD pipeline. Must be private.                                                                                                                |

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| <b>Application Load Balancer (ALB) / Ingress</b> | <b>AWS US-East-1</b> | L7 ingress in front of the container services. Terminates TLS (or passes through), routes to services, supports access logs, and provides a clear attachment point for AWS WAF if Cloudflare is bypassed or used DNS-only. |
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| <b>Identity &amp; Access</b> | <b>AWS IAM</b> | Role-based access control for services, CI/CD, and admins. Must enforce least privilege, MFA for console access, and use role assumption for deployments. (Do <b>not</b> require or transfer AWS Root credentials.) |
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| <b>Network Security</b> | <b>AWS VPN</b> | <p>All assets management interfaces must be secured (SSH, TELNET, EXEC, etc..)</p> <p>Access to these interfaces via shell or terminal will be allowed only by connection with AWS VPN</p> |
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| <b>VPC Security</b> | <b>AWS VPC w/ AWS VPN</b> | <p>VPN access should span multiple VPCs.</p> <p>Operations VPC to house operations dashboards and assets which will not have a public based interface. Routing to these assets should be restricted to VPN only</p> <p>Engineering VPC which will house all Stack Trading publicly facing technology. Web traffic allowed but management interface traffic restricted to VPN only</p> |
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| <b>Secrets Management</b> | <b>AWS Secrets Manager (or SSM Parameter Store + KMS)</b> | Secure storage/rotation of API credentials (Rithmic/MT5/TraderEvolution/YourBourse/ Payment Providers). No secrets in GitHub Actions plaintext. Must support rotation and environment separation (dev/stage/prod). |
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| <b>Container Registry Management</b> | <b>Amazon ECR (or similar alternative)</b> | Storage of container images for middleware, bot, workers, wrappers. Supports vulnerability scanning and deployment automation. |
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| <b>Email / Notifications</b> | <b>Amazon SES (or equivalent transactional email provider)</b> | Transactional emails (verification, password reset, alerts) and system notifications. Must support SPF/DKIM/DMARC and environment isolation. |
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| <b>Security Audit</b> | <b>AWS CloudTrail (Account-level /</b> | <b>Immutable audit trail</b> of AWS asset |
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| <b>Logging</b> | <b>Org-level)</b> | activity (console + API calls + HTTP traffic) across all regions. Required for incident response, regulatory traceability, and change accountability. <b>Connectivity:</b> Delivers logs to a dedicated <b>S3 bucket</b> (preferably Object Lock/WORM) and/or centralized logging destination. |
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| <b>Threat Detection</b> | <b>Amazon GuardDuty or similar</b> | <b>Managed threat detection</b> for anomalous activity across AWS accounts (IAM anomalies, unusual API calls, suspected credential compromise, suspicious network behavior). <b>Connectivity:</b> Feeds findings to <b>Security Hub</b> and alerting pipeline (Ops Slack / ticketing / incident workflow). |
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| <b>Security Posture Management</b> | <b>AWS Security Hub</b> | <b>Centralized security findings and compliance posture</b> aggregator across supported AWS services (including GuardDuty findings). Enables ongoing visibility and baseline controls. <b>Connectivity:</b> Integrates with alerting/ticketing and reporting; supports multi-account rollup if AWS Organizations is used. |
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| <b>Private Connectivity Resilience</b> | <b>AWS Direct Connect + Site-to-Site VPN Failover</b> | Ensures AWS ↔ NY4 control-plane connectivity remains available if the primary path degrades. <b>Connectivity:</b> Primary path via <b>Direct Connect</b> ; secondary path via <b>AWS Site-to-Site VPN</b> (automatic or documented manual failover). Vendor must document the failover procedure, monitoring, and |
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|  |  | operational runbooks. |
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| <b>Cost Controls &amp; Allocation</b> | <b>AWS Budgets + Cost Explorer + Tagging / Cost Categories</b> | Implements <b>cost guardrails</b> and spend visibility across environments (dev/stage/prod) and components (API, DB, logs, networking). <b>Connectivity:</b> Budget alerts route to finance/ops channels; mandatory tagging policy (e.g., <b>env, service, owner, cost_center</b> ) enables showback/chargeback and faster optimization. |
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## 2.2 Execution Gateways

The Middleware Hub functions as the administrator, pushing risk parameters to the Execution Cores where they are enforced locally. The vendor must construct a unified "Execution Abstraction Layer" that standardizes commands across physically distributed gateways. The Middleware Hub (AWS) pushes risk parameters to these gateways, which enforce them via the respective institutional APIs.

**A. Futures Gateways (Hosted in AWS US-East-1)** The execution wrappers for Futures markets must be deployed as containerized services within the Core Compute cluster. They communicate with the Exchange Hubs (Chicago) via the AWS high-speed public backbone.

- **Rithmic R+:** Integration via R | API+. The Daily Loss Limit and Level Stop Loss values must be loaded directly into the Rithmic Exchange execution engine (server-side) to ensuring auto-liquidation occurs even if the user's client disconnects.
- **Rithmic Gateways Bifurcation:** Agency A must construct routing logic that isolates simulated flow from live flow using two distinct API environments provided by Rithmic.
  - Level 0 (Associate Track): All simulated evaluation accounts must be provisioned exclusively on the Rithmic Paper Trading Gateway. This environment is isolated from the Futures Commission Merchant to prevent incurring live routing fees.
  - Level 1 and Above (Live DMA): All live funded accounts must be provisioned on the Live FCM Rithmic Gateway.

**B. Forex Gateway (Hosted in Equinix NY4)** The Forex execution logic resides on the dedicated Windows Server in NY4. The Middleware Hub connects to this gateway via an AWS Direct Connect private fiber link to issue commands with zero internet jitter.

- **MetaTrader 5 (MT5):** Integration via MT5 Manager API to automate account provisioning, password resets, and real-time equity monitoring.
- **TraderEvolution Main Label:** Integration via the TraderEvolution Admin API to handle account provisioning, real-time equity monitoring, and natively enforce Profit Targets and Stop Losses in server RAM.
- **BourseYours Bridge:**

Abri Van Rooyen will be responsible for configuring the FIX routing logic within the YourBourse environment to ensure trades map correctly from both TraderEvolution and MT5 to the Liquidity Providers. Market entry orders route based on lot size. Orders totaling less than **0.34** standard lot route to Forex.com, while orders of **0.34** standard lot or greater route to Interactive Brokers (**StoneX negotiation: [Forex.com](#) is \$12/lot and IBKR is \$2/lot with \$2 minimum**). YourBourse acts as the central translation engine holding the ledger. When inbound fills arrive from the Liquidity Providers via the Stack Trading Omnibus corporate account, the bridge reads the FIX Client Order ID to reverse map the fill and route the confirmation exclusively to the originating sub account on the exact originating platform, which will update the specific profit and loss, margin, and open positions for that individual trader.

- **MT5 Infrastructure:** MT5 natively handles Stop Loss parameters. However, MT5 does not natively support an automated Profit Target trigger to flatten positions and lock the account. To handle this, a server-side plugin (YourBourse Contest Stop Out) will be utilized on the MT5 White Label server.
- **MT5/Plugin Vendor Responsibility:** The White Label provider's DevOps team handles the physical installation, configuration, and associated setup fees for this plugin on the NY4 server.
- **Agency A Scope:** Agency A is responsible for integrating with the plugin's API.
- **Execution Flow:** Upon account creation, level promotion, or account reset, the Node.js middleware must execute a POST request to the plugin API to set the exact Profit Target for that specific MT5 login. When the plugin triggers (flattening positions and locking the account upon reaching the target), the backend must detect this state change and fire the appropriate webhook to initiate the Level-Up activities and provision the new risk limits and notional values.
- **Client Interfaces (Forex):** Phase 1 includes multiple client interfaces (e.g., TradingView Terminal and native TraderEvolution UI) that connect to the TraderEvolution Main Label execution engine. The vendor must clarify in the proposal which interface deployments/domain routing/SSO components are included in their scope versus handled by the TraderEvolution/white-label provider, and what

integration dependencies exist for end-to-end testing.

## 2.3 Discord Bot & Message Queueing

The system requires a dedicated Node.js Discord Bot application operating alongside the primary API middleware. Discord imposes rate limits for posting messages (5 messages per 5 seconds per channel). Look at discord.js to reduce overhead and more easily manage. Do not use axios due to recent malicious exposure.

- **The Queue:** All messages originating from the Send Broadcast Chat dashboard endpoint must be pushed into a Redis memory queue, not sent directly to Discord.
- **The Batch Worker:** A worker process must consume the Redis queue. If inbound velocity exceeds 1 message per second, the worker must concatenate multiple user messages into a single Discord API payload ( [User A]: Message \n [User B]: Message) to respect the rate limit while maintaining near-instant delivery to the live-broadcast channel.
- **Discord Application Commands (Slash Commands)**
  - **The Command:** /submit-bounty [url]
  - **Validation:** The bot must perform a client-side check to ensure the input string is a valid TikTok or YouTube Shorts URL before accepting the submission.
  - **Channel Permissions:** Channel Permissions: The #bounty-submissions channel must have Send Messages disabled for the @everyone role, but Use Application Commands enabled. This forces a structured input environment.
  - **Rate Limiting:** The bot must enforce a one submission per 24 hours rule per discord\_user\_id to prevent spamming the AI verification engine.
  - **Automated Support Interception & Routing**
    - Monitor: The Discord Bot must actively monitor #the-pit and #verify for operational complaints (e.g. slippage, lag, frozen platform).
    - LLM Sentiment Gate: To prevent false positives (e.g. "Stack Trading has no slippage!"), the bot must route messages containing trigger words to a lightweight Gemini LLM prompt to classify the intent as either a 'Complaint' or 'Positive/Neutral'. This is the same light LLM we need anyway for "The Desk Manager" commentary.
    - Execution: If classified as a Complaint, the bot must instantly delete the message, reply with a public placeholder tagging the user ("@[User], to protect your account details and resolve this immediately, we have opened a secure Zendesk support ticket for you. Please check your registered email."), and fire a webhook to Zapier containing the user's discord\_user\_id and the original message text.

## 3.0 Detailed Scope of Work

Agency A is responsible for the end-to-end backend architecture and API integration. Frontend design (Agency B) is excluded.

### 3.1 Primary Deliverables

1. **Prop Tech Hub:** A middleware application hosted on AWS. Preferred implementation is **Node.js/TypeScript** (to align with existing design assumptions), however vendors may propose an alternative backend stack (e.g., Go/Java/.NET) if they can justify performance, maintainability, hiring/ops implications, and preserve the required OpenAPI + event contracts.
2. **Database & Storage:** Normalized Amazon RDS schema and Amazon S3 buckets.
3. **Gateway/API Wrappers:** Gateway/API Wrappers: Custom connectors and adapters for **Rithmic (Futures), MT5 Manager API (Forex), TraderEvolution Admin APIs**, and integration touchpoints to the **YourBourse Bridge** (routing/translation context), including integration support for the MT5 profit-target/contest plugin API.
4. **Operational Logic Flows:** Implementation of Operational Flows (detailed in the attached *Zapier* document).
5. **API Stubs:** Static API endpoints delivered early in the project to unblock the frontend development team.
6. **Discord Bot:** A custom, persistent Discord Bot (hosted on AWS) is required. This is necessary to handle Digital Eviction (reliably stripping roles upon termination) and Pod Management (programmatically creating private channels and managing user permissions), which exceeds standard webhook capabilities.
7. **Immutable Ledger (Smart Contract):** A single Solidity smart contract deployed to the Polygon PoS Mainnet and the associated Node.js middleware to write generalized corporate metrics to the public blockchain transaction logs.
8. Configuration of **AWS CloudWatch Dashboards** to visualize system health (API Latency, Error Rates, Webhook Failures, Database CPU).
9. Configuration of **CloudWatch Alarms** for Severity 1 events (e.g. "Middleware Hub Unresponsive," "Rithmic Disconnect") to trigger immediate alerts to the Ops Slack channel.
10. **Version Control:** Private source control repository (GitHub, GitLab, or equivalent) with automated CI/CD (GitHub Actions, GitLab CI, or AWS-native pipeline) to build, scan, and deploy infrastructure and services.
11. **Security Controls:** Setup of IAM roles, Network Security for assets and VPCs, VPN, Monitoring, Threat Detection, SAST/DAST scanning, and application logging.

### 3.2 Calculations and Endpoints for our Dashboard

You are responsible for the calculation logic and API endpoints for the following features. The

Dashboard Agency (Agency B) will only visualize the JSON data you provide.

### Real-Time Data Streaming (WebSocket):

The dashboard requires high-fidelity streaming for account metrics and immediate notification of binary state changes (e.g. Promotion or Failure) to trigger UI animations and lockouts without polling.

| Event Channel      | Trigger Condition                                                                                                                       | Payload Structure                                                                                                                                                                                                                                                                                                                                                                                                                                            | Frontend Behavior                                                                                                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACCOUNT_METRICS    | <b>Continuous:</b><br>Throttled to 1 evaluation/push per second (1000ms). from Execution Core (Rithmic/MT5/TraderEvolution).            | JSON Object {<br>"balance": float,<br>"equity": float,<br>"open_pnl": float,<br>"realized_pnl": float }                                                                                                                                                                                                                                                                                                                                                      | Update Header:<br>Updates the primary "Account Balance" and "Equity" displays in real-time.                                                                                                                                   |
| POSITION_METRICS   | Continuous: Streams whenever an order is filled, partially filled, or closed.<br><br>Throttled to 1 evaluation/push per second (1000ms) | JSON Array of Objects {<br>"symbol": string, "side": "Long/Short", "net_pos": integer, "unrealized_pnl": float, "realized_pnl": float, "total_pnl": float, "est_swap": float }                                                                                                                                                                                                                                                                               | Update Table: Re-renders the Position Monitor rows and recalculates the bottom totals row.                                                                                                                                    |
| PERFORMANCE_UPDATE | Middleware Trigger: Fires upon the closure of a trade.                                                                                  | JSON Object {<br>"sharpe_ratio": float,<br>"sortino_ratio": float,<br>"profit_factor": float,<br>"win_rate_percent": float,<br>"avg_win": float,<br>"avg_loss": float,<br>"latest_trade": { "pnl": float, "duration_sec": integer, "trade_efficiency_percent": float, "risk_reward": float, "is_winner": boolean },<br>"holding_time": {<br>"avg_win_min": float,<br>"avg_loss_min": float,<br>"risk_mitigation_text": string, "wins_distribution": [[x,y]], | Update Dashboard:<br>Smoothly updates all performance gauges. Appends the latest_trade object as a dot on the Trade Efficiency Matrix and Win Rate grid. Morphs the Holding Time bell curves to the distribution coordinates. |

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|                           |                                                                              | "losses_distribution": [[x,y]]<br>}}                                                                                                                               |                                                                                                                                                                                                                                                        |
| <b>RULE_TRACKING</b>      | Continuous: Real-time calculation of distance to hard limits.                | JSON Object {<br>"dist_daily_loss": float,<br>"dist_level_stop": float,<br>"current_daily_pnl": float,<br>"daily_loss_limit": float }                              | Update Gauge: Updates the "Daily Loss Limit Remaining" visual and "Drawdown Heatmap."                                                                                                                                                                  |
| <b>LEVEL_PROMOTION</b>    | <b>Flow 4 Trigger:</b> When Current_Level increments in the database.        | JSON Object { "event": "PROMOTION",<br>"old_level": int,<br>"new_level": int,<br>"timestamp": "ISO8601"<br>}                                                       | Trigger Celebration: Immediately plays the "Level Up" animation/confetti and unlocks the next level on the Bento Grid.                                                                                                                                 |
| <b>DAILY_LIMIT_BREACH</b> | <b>Real-Time Monitor:</b><br>When Net_P&L <= Daily_Loss_Limit.               | JSON Object { "event": "DAILY_STOP",<br>"current_loss": float,<br>"limit": float,<br>"reset_time": "16:00 CST" }                                                   | Trigger Soft Lock: Displays "Daily Limit Reached" overlay. Disables "Enter Order" buttons until the reset_time                                                                                                                                         |
| <b>LEVEL_STOP_BREACH</b>  | Equity <= Market_Loss_When_Stopped                                           | JSON { "event_type": "LEVEL_STOP_BREACH",<br>"user_id": "[user_id]",<br>"equity": float,<br>"market_loss_when_stopped": float,<br>"amount_over_stop_loss": float } | Trigger Hard Lock: Displays "Account Failed" (or "Defense Mode" if eligible) overlay.<br><br>Triggers Flow 7 (Termination) or Flow 4B (Defense). Pushed immediately upon Hard Breach.<br>amount_over_stop_loss =<br>Market_Loss_When_Stopped - Equity. |
| <b>NOTIFICATION</b>       | Zapier Trigger: Any global alert from Flow 23 (News) or Flow 24 (Overnight). | JSON Object { "severity": "INFO/WARN/CRITICAL",<br>"message": "Market Close in 10m", "timestamp":                                                                  | Trigger Notification Center: Displays a persistent notification. Notifications ordered by timestamp                                                                                                                                                    |

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|                             |                                                                                       | "ISO8601",<br>"target_timestamp":<br>"ISO8601" }                                            | descending.                                                                                                                                             |
| <b>MARKET_STATUS_UPDATE</b> | Zapier Trigger: Flow 24 (Overnight/Margin Milestones).                                | JSON Object {<br>"asset_class": "string",<br>"state": "string",<br>"short_text": "string" } | Update Dashboard Widget:<br>Instantly updates the Market Status Widget (top right) to display the current state and the short countdown or status text. |
| <b>DISCORD_CHAT_STREAM</b>  | Continuous: Whenever a new message is posted in the active Discord broadcast channel. | JSON Object { "author": "string",<br>"message_text": "string", "timestamp": "ISO8601" }     | Append to Chat:<br>Instantly appends the read-only message to the bottom of the Live Chat Feed UI component.                                            |
| <b>GEKKO_CHATTER</b>        | Asynchronous LLM Trigger (The Desk Manager)                                           | JSON { "text": "string",<br>"vibe": "string",<br>"event_type": "string" }                   | Renders "The Desk Manager" speech text                                                                                                                  |

### 3.2.1 The Desk Manager (LLM Commentary Engine)

The Middleware Hub must implement a stateless commentary engine utilizing the **Google Gemini 1.5 Flash 8B API**. This engine operates as an asynchronous worker to generate dynamic, unique commentary based on the logic below.

#### A. Event Filtering (The Notable Events)

The engine triggers **ONLY** on the following events to ensure maximum impact and cost-efficiency:

- **SESSION\_START**: First dashboard login of the day (UTC 00:00 reset).
- **LOSS\_EVENT**: A single loss exceeding 1% of virtual equity or a third consecutive losing trade in a session.
- **PROMOTION**: Profit target reached or Level promotion.
- **DEFENSE\_TRIGGERED**: When a trader enters the Drawdown Defense protocol (Flow 4B).
- **STOP\_HIT**: When a Daily Stop or absolute Level Stop is reached (Trader is "Fired").
- **NEWS\_HOLD**: Holding a position during a High-Impact News event (Red Folder) regardless of whether a rule breach occurred.
- **ORDER\_CANCELLED**: A streak of 5+ cancelled orders within a 10-minute window (The Hesitation Check).

- **POD\_DYNAMICS:** A Pod Leader receiving a bonus or a Pod Member being promoted.
- **PAPER\_HANDS:** Closing a winning trade where less than 10% of the level's Profit Target was realized (Lack of conviction).
- **STAGNATION:** 72 hours of zero trading activity during an active Associate Track challenge.
- **THE\_WHALE:** Opening a position size that utilizes more than 80% of the allowed margin for the current account level.
- **REVENGE\_TRADE:** Immediate re-entry (within 60s) following a loss, specifically if the position size is increased.
- **THE\_CLOCK\_WATCHER:** Closing a trade within 60 seconds of the market close or overnight session break.
- **SCALP\_HUNTER:** Executing 10+ round turns within a 5-minute window (Over-trading check).
- **COMMUNITY\_LIVE:** Triggered globally via webhook when the official Stack Trading YouTube stream goes live.

## B. Rate Limiting

- **Algorithm:** Token Bucket.
- **Limit:** 10 critiques per hour per user.

## C. Implementation Specification: The Master Prompt Payload

The backend must pass the following structured system instruction and JSON context to the Gemini API.

### System Instruction:

"You are Gordon Gekko, a ruthless, cynical 1980s Wall Street raider. You are analyzing the trades of a junior trader using YOUR capital. Incompetence is a personal insult. Remind the trader this is real capital and a professional firm, not a casino video game. Critique the move in under 30 words using 80s jargon like pal, rookie, or liquidated. Logic constraints:

1. Respect Scale: Levels 1-5 get maximum disdain. Respect increases from Level 6 (W2) up to Level 24 (peer respect).
2. If event\_type is NEWS\_HOLD, ask why they are gambling with your liquidity during red folders.
3. If event\_type is STOP\_HIT, be the Executioner. Tell them to get off your floor.
4. If event\_type is DEFENSE\_TRIGGERED, act as the Warden who pulled the plug to save his money.
5. If event\_type is PAPER\_HANDS, mock their lack of conviction and 'poverty mindset.'
6. If event\_type is THE\_WHALE, mock their arrogance for over-leveraging your capital.
7. If event\_type is REVENGE\_TRADE, call out their emotional instability and gambling.
8. Pod Role: If Leader, acknowledge their responsibility for the group. If Member, mock their reliance on the leader.
9. P&L: For wins, provide backhanded, cynical compliments. For losses, be biting but brief.

10. If pulse\_score < 50, prioritize attacking their lack of discipline (averaging down) over P&L.
11. Use the variability\_seed and Temperature 0.9 to ensure no two responses are the same. Return ONLY the raw string of the commentary."
12. If event\_type is THE\_CLOCK\_WATCHER, mock them for being terrified of the overnight session.
13. If event\_type is SCALP\_HUNTER, ask if they're trying to set a record for commissions or actually trade.
14. If consecutive\_losses > 3, mock their inability to walk away and tell them they are funding your yacht.
15. If hold\_time\_seconds < 15 on a loss, mock their panic and weak execution. If hold\_time\_seconds > 3600 on a loss, mock them for "marrying" a bad position.
16. If distance\_to\_daily\_stop\_pct < 10%, adopt the "Executioner" vibe. Warn them that you have your finger on the liquidation button.
17. If event\_type is COMMUNITY\_LIVE, cynically order them to navigate to the Community Page to watch the live stream so they can finally learn how to trade properly.
18. If asset\_class is FOREX, cynically refer to them as a retail pip-chaser or currency tourist. If FUTURES, mock them for getting crushed by the real institutional order book.

### JSON Context Structure:

```
{
  "user_context": {
    "first_name": "string",
    "current_level": "integer (0-24)",
    "pod_role": "ENUM [LEADER, MEMBER, NONE]",
    "asset_class": "ENUM [FUTURES, FOREX]",
    "days_remaining": "integer"
  },
  "trade_context": {
    "event_type": "ENUM [TRADE_CLOSED, NEWS_HOLD, ORDER_CANCELLED, SESSION_START, DEFENSE_TRIGGERED, STOP_HIT, PAPER_HANDS, STAGNATION, THE_WHALE, REVENGE_TRADE, THE_CLOCK_WATCHER, SCALP_HUNTER, COMMUNITY_LIVE]",
    "symbol": "string",
    "side": "ENUM [LONG, SHORT, NONE]",
    "pnl": "float",
    "hold_time_seconds": "integer",
    "consecutive_losses": "integer"
  },
  "risk_state": {
    "pulse_score": "integer (0-100)",
    "distance_to_daily_stop_pct": "float",
    "distance_to_max_drawdown_pct": "float"
  },
  "widget_metrics": {
```

```

"sharpe_ratio": "float",
"win_rate": "float"
},
"system": {
  "variability_seed": "random_64bit_int"
}
}

```

Note: pod\_role is dynamically evaluated by the Middleware Hub at runtime. If pod\_leader\_id is populated in the user's record = MEMBER. If the user's ID exists in the Pod\_Leader\_Registry = LEADER. Otherwise = NONE.

## Outbound Webhook Events:

Zapier acts as the listener. The Middleware Hub must push these events to trigger Zapier **Flow 4** and **Flow 7**.

| Event Type               | Trigger Condition                  | Payload Structure                                                                                                                                                  | Description                                                                                                                                                             |
|--------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PROFIT_TARGET_HIT</b> | Net Equity >= Profit Target        | JSON { "event_type": "PROFIT_TARGET_HIT",<br>"user_id": "[user_id]",<br>"net_profit": float }                                                                      | <b>Triggers Flow 4 (Promotion).</b><br><br>Pushed immediately when the target is reached (Real-time).                                                                   |
| <b>LEVEL_STOP_BREACH</b> | Equity <= Market_Loss_When_Stopped | JSON { "event_type": "LEVEL_STOP_BREACH",<br>"user_id": "[user_id]",<br>"equity": float,<br>"market_loss_when_stopped": float,<br>"amount_over_stop_loss": float } | <b>Triggers Flow 7 (Termination) or Flow 4B (Defense).</b><br><br>Pushed immediately upon Hard Breach.<br>amount_over_stop_loss =<br>Market_Loss_When_Stopped - Equity. |

## Reference Data and User State Endpoints:

These endpoints handle static or low-frequency data retrieval (e.g. user profile, career map, and institutional swap rates). These should be fetched via standard HTTP requests on session load or on demand, rather than streamed via WebSocket.

| Endpoint                    | Method | Input Parameters | Output Structure                                                                                                                                                                                                                                             | Logic Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|--------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Current Level Detail</b> | GET    | user_id          | JSON Object (Level, Title, Notional, Target, Stop, Bonus, Salary, PTO, expiration_date, expiration_timestamp, days_remaining, defense_used, defense_attempt_count, platform_username, platform_password, license_key, connection_gateway, payout_portal_url) | <p>Returns the trader's current status. Used to populate the dashboard header, "Next Level" progress bars, and the PnL Gauge defense state. Calculate days_remaining Server-Side (UTC). Return an integer. If days_remaining &lt; 0, return 0.</p> <p>Payout Routing: Evaluate the user's Level and W-2 status.</p> <p>IF user is Level 1 through 5 (Global) OR Level 6+ (International Contractor): populate payout_portal_url with the static Dots login URL.</p> <p>IF user is Level 6+ (US W-2): populate payout_portal_url with the Rippling employee portal URL.</p> |
| <b>24 Levels Detail</b>     | GET    | None             | JSON Array of Objects (Level 1-24 details)                                                                                                                                                                                                                   | Returns the full configuration matrix (Levels 1-24) from the Zapier Tables/DB source. Used to render the "Bento Grid" career map.                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Platform Credentials</b> | GET    | user_id          | JSON Object (Platform, Login, Connection_String, Connection_Guide_URL)                                                                                                                                                                                       | Secure: Returns the specific login credentials and connection string for the user's chosen execution platform (Rithmic/MT5/TraderEvolution) so they can connect the terminal. The middleware evaluates platform_selection and returns the specific PDF or Notion guide link.                                                                                                                                                                                                                                                                                               |
| <b>HR Integration</b>       | GET    | user_id          | JSON Object {<br>"vacation_days": int,<br>"next_pay_date": "YYYY-MM-DD",<br>"base_salary_annualized": float,<br>"benefits_estimate":                                                                                                                         | Server-side Integration: Reads compensation_track from Users table. If User is W2: Query the Rippling API to fetch real-time HR data for W-2 traders. If User is Level 6 or higher contractor: Query the SQL Database for the retainer amount.                                                                                                                                                                                                                                                                                                                             |

|                                      |      |                                  |                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------|------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      |      |                                  | float,<br>"compensation_tracking": string }                                                                                                             |                                                                                                                                                                                                                                                                                                                                            |
| <b>Certificate</b>                   | GET  | user_id                          | JSON Object { "url": string,<br>"polygon_tx_hash": string,<br>"promotion_date": "ISO8601",<br>"bonus_awarded": float,<br>"capital_allocation": string } | Returns the raw data required for the frontend to dynamically render the Certificate Modal, including the Polygon transaction hash for the QR code, alongside the legacy S3 URL for PDF downloads.                                                                                                                                         |
| <b>Swap Rates</b>                    | GET  | symbol (String, Optional)        | JSON Object {<br>"symbol": string,<br>"swap_long": float,<br>"swap_short": float }                                                                      | Retrieve current daily Swap Rates from the Liquidity Provider (BourseYours/MT5). Used to populate the "Overnight Fees" information modal before market close.                                                                                                                                                                              |
| <b>Resign Account</b><br><br>/resign | POST | User_id,<br>confirmation_phrase  | HTTP 200 OK                                                                                                                                             | Trigger Flow 20: Immediately sets account to "ReadOnly", cancels open orders, flattens positions, and logs the user out. This action is irreversible.                                                                                                                                                                                      |
| <b>AI Preference Toggle</b>          | POST | user_id,<br>ai_enabled (Boolean) | HTTP 200 OK                                                                                                                                             | Updates the ai_enabled flag in the user's profile. Used by the frontend to turn the "Risk Personality" coach on or off.                                                                                                                                                                                                                    |
| <b>Market Data Products</b>          | GET  | user_id                          | JSON {<br>"products": [ {<br>"id":<br>"NYMEX_Bundle",<br>"name": "NYMEX Data", "price": 115.00 } ] }                                                    | <ul style="list-style-type: none"> <li>• Source: Fetch current products and pricing from Zapier Table C (Global Variables). Do not hardcode prices.</li> <li>• Status: Return only products the user does not currently have active (User State check).</li> <li>• Usage: Populates the "Manage Add-ons" and "Checkout" modals.</li> </ul> |
| <b>Public Market</b>                 | GET  | None                             | JSON {<br>"products": [ {                                                                                                                               | <b>Public Pricing:</b> <ul style="list-style-type: none"> <li>• Return list of <i>all</i> available Add-ons</li> </ul>                                                                                                                                                                                                                     |

|                         |      |                             |                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|------|-----------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data                    |      |                             | "id": "ny_data",<br>"price": 115.00 } ]<br>}                   | and prices from Zapier Table C.<br><br>• <b>Difference:</b> Unlike the private endpoint, this does <i>not</i> filter by "what the user owns" ( user doesn't exist yet).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Public Platform Options | GET  | asset_class                 | JSON {<br>"platforms":<br>["NinjaTrader 8",<br>"Quantower" ] } | <b>Dynamic Registry Lookup:</b><br><br>• <b>Query:</b> Fetch rows from <b>Zapier Table I</b> where Asset_Class matches input AND Is_Active is TRUE.<br><br>• <b>Return:</b> Array of Platform_Name strings.<br><br>• <b>Purpose:</b> Populates the frontend dropdown. Allows Ops to hide a broken platform instantly via Zapier.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Auth Session Handshake  | POST | auth_provider_token (JWT)   | JSON Object {<br>"user_id": "UUID",<br>"role": "trader" }      | Validates the Firebase/Auth0 token from the frontend, maps it to the internal SQL User ID, and establishes the secure session for all subsequent API calls.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Process Veriff Handoff  | POST | provider_signature, payload | HTTP 200 OK                                                    | 1. Security: Validate the Veriff cryptographic signature.<br><br>2. Parsing: Extract user_id, verification_status, legal_name, and address from the Veriff payload. IF verification_status != 'approved', update Users table status and halt.<br><br>3. Existing Profile Check: Query PostgreSQL Users table for existing dots_payee_id. IF dots_payee_id is NOT NULL (Trader already exists in Dots), update local KYC status to 'Approved' and exit.<br><br>4. Profile Creation: IF dots_payee_id IS NULL, call Dots API (POST /payees) injecting the Veriff extracted legal_name and address to bypass manual data entry.<br><br>5. Database Anchor: Extract the returned Payee ID from the Dots response and save it to the dots_payee_id column in the Users |

|                       |      |                                                                                                                   |                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |      |                                                                                                                   |                                                                                                                             | <p>table.</p> <p>6. Trigger Next Step: Fire a webhook to Zapier to trigger the "Payout Profile Setup" email sequence containing the generated Dots white-labeled link (which natively collects the W-8BEN/W-9)..</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Calculate Cart</b> | POST | product_id,<br>addon_ids,<br>promo_code,<br>billing_country,<br>billing_region,<br>user_ip, user_id<br>(Optional) | <pre>JSON {   "base_price":   float,   "discount_amount   ": float,   "tax_amount":   float,   "total_price": float }</pre> | <p><b>1. Sanctions Gate (Critical):</b> Validate billing_country and billing_region against the Compliance_geo_restrictions PostgreSQL database table. Logic must explicitly allow traffic from nations with regional exemptions like Canada and Ukraine by evaluating the specific region parameter. IF a match is found where the database Country equals billing_country AND the database Region equals 'All' or equals the billing_region: Return HTTP 403 Forbidden immediately with the restricted Country and Region names in the payload.</p> <p><b>2. Pricing Engine:</b> IF product_id is 'EVAL_L1', 'EVAL_L2', or 'EVAL_L5': Query Table J (Evaluation and Reset Pricing) for the specific base price. IF Global_Var_Founder_Cohort_Open == TRUE, apply the locked Founder discount rate to the respective tier. For other products, fetch prices for product_id and addon_ids. Validate and apply promo_code. Math: Final_Amount = (Base + Addons) - Discount.</p> <p><b>3. Location Check:</b> If billing_country is missing, query Users table via user_id to retrieve it.</p> <p><b>4. Tax Engine:</b> Ping Quaderno API with user_ip, billing_country, and Final_Amount to retrieve the dynamically calculated tax_amount. (Quaderno handles domestic exemptions and international \$0 tax</p> |

|                         |      |                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |      |                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                               | <p>logic automatically).</p> <p>5. Return exact line items to frontend to populate the Order Summary or 1-Click Confirmation Modal.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Execute Checkout</b> | POST | product_id,<br>addon_ids,<br>promo_code,<br>billing_country,<br>billing_region,<br>user_ip,<br>payment_method,<br>payment_token (Optional),<br>timestamp_utc (Optional),<br>impact_click_id (Optional),<br>utm_source (Optional),<br>utm_medium (Optional),<br>utm_campaign (Optional),<br>utm_term (Optional),<br>utm_content (Optional), user_id (Optional) | JSON { "status": "Success" OR "Failed",<br>"error_code": "string",<br>"transaction_id": "string",<br>"gateway_payload": { } } | <p>1. Sanctions &amp; Pricing: Re-run Sanctions, Pricing, and Tax calculation server-side to ensure integrity.</p> <p>2. Audit: Generate a static PDF transaction receipt stamping the user_ip, timestamp_utc, and Payment Gateway Transaction ID. Upload the PDF to the AWS S3 Immutable storage (WORM compliant) and write the S3 URI to the user record.</p> <p>3. Routing Execution:</p> <p>IF "CC", "APPLE_PAY", or "GOOGLE_PAY", query active MIDs in Merchant_Routing_Config. Apply Weighted Round-Robin algorithm to select target MID. If payment_token is provided, ping NMI API to execute charge and extract customer_vault_id to save to PostgreSQL. If payment_token is omitted, retrieve existing customer_vault_id using user_id to execute vaulted charge. Update MID volume.</p> <p>IF "CRYPTO", ping Triple-A API to generate a secure session payload.</p> <p>IF payment_method contains "NOMUPAY", ping Nomupay API to initialize checkout session and return hosted pop-up URL.</p> <p>IF payment_method contains "EBANX", ping EBANX API to initialize checkout session and return hosted pop-up URL.</p> <p>IF payment_method contains "DUSUPAY",</p> |

|                            |      |                                                               |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------|------|---------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |      |                                                               |                                                | <p>ping Dusupay API to initialize checkout session and return hosted pop-up URL.</p> <p>IF "SKRILL", ping Skrill API to generate hosted checkout link.</p> <p>4. Attribution: Parse and write incoming parameters (utm_source, utm_medium, utm_campaign, utm_term, utm_content) and the impact_click_id directly into the core user database record.</p>                                                                             |
| <b>Issue Refund</b>        | POST | user_id, type ("MARKET_DATA" or "FULL")                       | JSON { "status": "Refunded", "amount": float } | <p>Query database to find original transaction ID and Gateway_Source. IF NMI: Ping NMI API to issue refund (covers CC, Apple Pay, and Google Pay). IF Triple-A: Ping Triple-A API to issue refund. IF NOMUPAY: Ping Nomupay API to issue refund. IF EBANX: Ping EBANX API to issue refund. IF DUSUPAY: Ping Dusupay API to issue refund. IF SKRILL: Ping Skrill API to issue refund. Update invoices table status to 'refunded'.</p> |
| <b>Claim Account</b>       | POST | email, transaction_id, new_password, phone_number, shirt_size | HTTP 200 + session_token                       | <p>Link the transaction_id to the email to verify payment authenticity. Store the phone_number and shirt_size (Tops: Hoodie, Shirt) in the user profile. Hash and set the new_password. Transition status from "Guest" to "Active_Sim".</p>                                                                                                                                                                                          |
| <b>Admin Impersonation</b> | POST | admin_token, target_user_id                                   | JSON Object { "impersonation_token": "JWT" }   | <p>Support Logic: Validates the requester has Admin role. Returns a specialized session token that allows the Dashboard to load data as if it were the target_user_id in Read-Only mode.</p> <p>The middleware must globally reject all 'Write' requests (POST/PUT/DELETE) when an <b>impersonation_token</b> is used, regardless of UI state.</p>                                                                                   |
| <b>Get Service Pricing</b> | GET  | user_id                                                       | JSON { "reset_fee": float, "extension_fee":    | <p>1. Dynamic Retrieval: Queries <b>Table J</b> (Reset Pricing Matrix) based on the user's evaluation track (Level 1, 2, or 5)</p>                                                                                                                                                                                                                                                                                                   |

|                                                                       |      |                                                                             |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------|------|-----------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                       |      |                                                                             | float, "currency": "USD" }          | <p>and Asset_Class to return the precise <b>reset_fee</b>.</p> <p>2. Fetches <b>extension_fee</b> from <b>Table J</b> for that specific tier.</p> <p>3. Used to dynamically populate the 1-Click Reset and Extension dashboard buttons.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Purchase Time Extension</b></p> <p>/purchase-time-extension</p> | POST | <p>user_id,<br/>payment_token (Optional),<br/>payment_method (Optional)</p> | <p>JSON { "status": "Success" }</p> | <p>1. Validation: Check Status == 'Active_Challenge'. Fetch Base Fee.</p> <p>2. Sanctions Gate (Critical): Query the Users table via user_id to retrieve the stored billing_country and billing_region. Validate these against the Compliance_geo_restrictions PostgreSQL database table. Logic must explicitly allow traffic from nations with regional exemptions like Canada and Ukraine by evaluating the specific region parameter. IF a match is found where the database Country equals billing_country AND the database Region equals 'All' or equals the billing_region: Return HTTP 403 Forbidden immediately with the restricted Country and Region names in the payload.</p> <p>3. Tax Engine: Ping Quaderno API with billing_country and Base Fee to dynamically calculate required tax.</p> <p>4. Math: Final_Charge = Base Fee + Tax.</p> <p>5. Execution and UI Fallback: Check Users table for an existing customer_vault_id (NMI). IF a vaulted card exists, execute NMI server-to-server charge via selected MID for the Final_Charge. IF NO vaulted</p> |

|                                                                         |      |                                                                                     |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                         |      |                                                                                     |                                         | <p>card exists (user paid via Crypto, Nomupay, EBANX, Dusupay, etc.), return a payload directing the frontend to jump to the standard Secure Checkout screen presenting the region-based options matching GET /system/status.</p> <p>6. Update expiration date (if NMI charge successful).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p><b>Purchase Challenge Reset</b></p> <p>/purchase-challenge-reset</p> | POST | <p>user_id,<br/>payment_token<br/>(Optional),<br/>payment_method<br/>(Optional)</p> | <p>JSON { "status":<br/>"Success" }</p> | <p><b>1. Validation:</b> Verify Status == 'Failed' AND user is in an Evaluation Track (not live funded).</p> <p><b>2. Pricing:</b> Query Users for user_id. If is_founder is TRUE, use grandfathered reset fee. If FALSE, query <b>Table J</b> using the user's current_level and asset_class to set the Base Fee.</p> <p><b>3. Sanctions Gate (Critical):</b> Query the Users table via user_id to retrieve the stored billing_country and billing_region. Validate these against the Compliance_geo_restrictions PostgreSQL database table. Logic must explicitly allow traffic from nations with regional exemptions like Canada and Ukraine by evaluating the specific region parameter. IF a match is found where the database Country equals billing_country AND the database Region equals 'All' or equals the billing_region: Return HTTP 403 Forbidden immediately with the restricted Country and Region names in the payload.</p> <p><b>4. Tax Engine:</b> Ping Quaderno API with billing_country and Base Fee to dynamically calculate required tax.</p> <p><b>5. Math:</b> Final_Charge = Base Fee + Tax.</p> <p><b>6. Execution and UI Fallback:</b> Check Users table for an existing customer_vault_id (NMI). IF a vaulted card exists, execute NMI server-to-server charge via selected</p> |

|                                                                 |      |                                                                          |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------|------|--------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                 |      |                                                                          |                                                                            | <p>MID for the Final_Charge. IF NO vaulted card exists (user paid via Crypto, Nomupay, EBANX, Dusupay, etc.), return a payload directing the frontend to jump to the standard Secure Checkout screen presenting the region-based options matching GET /system/status.</p> <p>7. Reset: Upon successful NMI charge, execute an atomic server-side reset. Wipe the failed state, clear trade history, and reset the virtual balance back to the exact starting capital of that specific tier (Level 1, 2, or 5) on the simulation gateway.</p>                                                                                                                                                                                                                                                                                                                                                     |
| <b>Get Failure State</b>                                        | GET  | user_id                                                                  | JSON Object {<br>"is_breached":<br>bool,<br>"archive_date":<br>"ISO8601" } | <p>State Check: If status == 'Failed', return true and the scheduled data archival date (Failure + 10 Days).</p> <p>Used to trigger the "Risk Parameter Breached" overlay and the countdown timer on the Dashboard.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Initiate Re-Buy (Challenge Retry)</b><br><br>/initiate-rebuy | POST | user_id,<br>payment_token<br>(Optional),<br>payment_method<br>(Optional) | JSON { "status":<br>"Success" }                                            | <p>1. Validation: Check Status == 'Failed' OR Status == 'Terminated' on a Live DMA account profile to block active funded traders from abusing the link.</p> <p>2. Pricing Logic: If is_founder is TRUE, use their grandfathered locked_rebuy_price. If FALSE, query Table J to fetch the base Challenge Price corresponding to the user's active tier (Level 1, 2, or 5). Math: Discounted_Price = Challenge_Price * (1 - Retry_Discount).</p> <p>3. Sanctions Gate (Critical): Query the Users table via user_id to retrieve the stored billing_country and billing_region. Validate these against the Compliance_geo_restrictions PostgreSQL database table. Logic must explicitly allow traffic from nations with regional exemptions like Canada and Ukraine by evaluating the specific region parameter. IF a match is found where the database Country equals billing_country AND the</p> |

|                                  |      |                                                        |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------|------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  |      |                                                        |                                                                                                     | <p>database Region equals 'All' or equals the billing_region: Return HTTP 403 Forbidden immediately with the restricted Country and Region names in the payload.</p> <p>4. Tax Engine: Ping Quaderno API with billing_country and Discounted_Price to dynamically calculate required tax.</p> <p>5. Math: Final_Charge = Discounted_Price + Tax.</p> <p>6. Execution and UI Fallback: Check Users table for an existing customer_vault_id (NMI). IF a vaulted card exists, execute NMI server-to-server charge via selected MID for the Final_Charge. IF NO vaulted card exists, return a payload directing the frontend to jump to the standard Secure Checkout screen presenting the region-based options matching GET /system/status.</p> <p>7. Provisioning: If transaction is successful, archive the current failed live ledger data. Permanently drop the live sub-account to stop routing fees, reset the user profile back to an evaluation track at their current tier (Level 1, 2, or 5), and call the standard Provision Simulation User logic to issue a fresh challenge on the isolated simulation gateway.</p> |
| <b>Provision Simulation User</b> | POST | user_email,<br>asset_class,<br>platform,<br>product_id | <pre>JSON { "status": "success", "username": "...", "password": "...", "license_key": "..."} </pre> | <p>1. Gateway Routing: Lookup Zapier Table I for Platform_Name. Route: IF Rithmic -&gt; Call Rithmic Paper Trading API Wrapper. IF MT5 -&gt; Call MT5 Manager API.</p> <p>2. Dynamic Template Mapping: IF asset_class == 'Futures', evaluate the incoming product_id. IF product_id is 'EVAL_L1', assign Rithmic_Stack_Template_Level_1_Eval. IF product_id is 'EVAL_L2', assign Rithmic_Stack_Template_Level_2_Eval. IF product_id is 'EVAL_L5', assign Rithmic_Stack_Template_Level_5_Eval.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                                                                    |      |                                                                                                   |                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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|                                                                    |      |                                                                                                   |                                                                                                                                                   | <p>3. Action: Create simulation user account and inject the corresponding mapped template parameters (starting balance, 14% target, 7.5% stop) directly into the gateway core.</p> <p>4. Platform Entitlement (NinjaTrader): IF platform == "NinjaTrader 8", simultaneously call the NinjaTrader Enterprise Prop API to provision the "Multi-Broker" entitlement using the user_email.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p><b>Provision Live User</b></p> <p>/provisioning/live-trader</p> | POST | <p>user_id,<br/>risk_group ("Live_W2"),<br/>data_selections,<br/>fcm_subaccount_id (Optional)</p> | <pre>JSON {   "live_username": "[username]",   "live_password": "[password]",   "license_key": "[license_key]",   "is_professional": true }</pre> | <p>Segregated Setup (Flow 3):</p> <ol style="list-style-type: none"> <li>1. Gateway Routing: Route to discrete Ironbeam sub accounts (Futures) or the Stack Trading Omnibus corporate account at <a href="https://forex.com/interactive-brokers">Forex.com/Interactive Brokers</a> (Forex) based on Asset_Class.</li> <li>2. Sub Account Creation: For Futures (Rithmic), provision the specific fcm_subaccount_id passed in the Zapier payload and assign to the predefined Risk Group Template (Rithmic). For Forex (MT5/TraderEvolution), create Sub Account ID: ST_LIVE_[ID] and assign to the TraderEvolution Group or MT5 Group to establish symbol permissions, position limits, and auto liquidate parameters for both maximum loss and target profit freeze. Capital Injection: Dynamically push the Level_Starting_Capital as the Account Balance parameter directly to the Sub Account via the respective Administration API.</li> <li>3. Professional Status Forced (Futures Only): Override any previous data selections and hardcode CME Subscriber Status to 'Professional' for all Rithmic Live accounts. 100% of live traders are classified as Pro.</li> <li>4. Risk Calculation &amp; Buffer: NinjaTrader</li> </ol> |

|                          |      |                                                 |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|                          |      |                                                 |             | <p>Exception: IF platform == "NinjaTrader 8", simultaneously call the NinjaTrader Enterprise Prop API to provision the "Multi-Broker" entitlement using the user_email. Return a blank string for license_key.</p> <p>5. Ledger Initialization: Insert: ledgers table. virtual_equity = [Level_Starting_Capital]. allocated_capital = [Level_Starting_Capital]. mapped_account_id = [Specific FCM Sub Account ID or Forex Omnibus Account ID].</p> <p>6. Return: Credentials (username, password, license_key).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Set Account State</b> | POST | user_id, state ("Active", "ReadOnly", "Locked") | HTTP 200 OK | <p><b>Flow 13 (DTO/Hiatus), Flow 14 (Disputes), &amp; Flow 4B (Drawdown Defense):</b></p> <ul style="list-style-type: none"> <li>• Execution Failsafe: Whenever state changes to 'Locked' or 'ReadOnly', the Node.js middleware MUST execute the corresponding API call to the execution server (Rithmic 'Suspend Trading', MT5 'Disable/Read-Only', or TraderEvolution 'Lock/Read-Only') to enforce the freeze at the exchange/broker level.</li> <li>• DTO/Hiatus Logic: Updates PostgreSQL to 'Locked' and suspends platform trading.</li> <li>• Dispute Logic: Updates PostgreSQL to 'ReadOnly' and suspends platform trading.</li> <li>• Drawdown Defense Logic: IF state == 'Locked' AND context == 'Defense', the Node.js middleware must natively initialize the Ethers.js wallet and execute contract.recordMetric() (metric_category="Risk Management Protocol", metric_value="Drawdown Defense Deployed (Seat Protected)", receipt_hash=[Internal Event ID]).</li> <li>• Termination Logic (Critical): IF state ==</li> </ul> |

|                                     |      |                                      |                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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|                                     |      |                                      |                                                                                              | <p>'Terminated', the middleware must execute a hard delete of the sub-account on the respective execution server (Rithmic Paper or Rithmic Live) via the DeleteUser API command to permanently stop the \$25 per user platform fee.</p> <p>IF platform == "NinjaTrader 8", the middleware must additionally call the NinjaTrader Enterprise Prop API to revoke the "Multi-Broker" entitlement and automatically return the seat to the firm's pool.</p> <ul style="list-style-type: none"> <li>• Re-Enable: Sets account back to Active and pushes the 'Restore Trading' API command to the execution server when DTO/Hiatus ends or Dispute resolves.</li> </ul> |
| <b>Toggle Market Data</b>           | POST | user_id,<br>enable_data<br>(Boolean) | HTTP 200 OK                                                                                  | <p><b>Flow 12 (Zombie) &amp; Flow 22 (Billing):</b></p> <ul style="list-style-type: none"> <li>• <b>Action:</b> Call Broker API (<b>Rithmic / MT5 / TraderEvolution</b>) to toggle real-time feeds.</li> <li>• <b>Usage:</b> Disables cost for Zombies; Downgrades to "Base Data" for unpaid bills.</li> </ul>                                                                                                                                                                                                                                                                                                                                                    |
| <b>Reset Platform 2FA</b>           | POST | user_id                              | HTTP 200 OK                                                                                  | <p><b>Flow 16 (Security Reset):</b></p> <ul style="list-style-type: none"> <li>• <b>Action:</b> Call Broker API (<b>Rithmic/MT5/TraderEvolution</b>) to invalidate 2FA tokens.</li> <li>• <b>Audit:</b> Log event to SQL</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Verify Payout Reconciliation</b> | POST | user_id                              | JSON {<br>"pnl_match": bool,<br>"broker_equity":<br>float,<br>"institution_balance": float } | <p>Flow 5 (Payout Safety):</p> <p>Reconciliation Only: Queries Sub Account Broker Equity (Rithmic/MT5/TraderEvolution). For Futures, validates against the mapped Ironbeam FCM Sub Account Balance. For Forex, due to size based order routing, validates against the aggregated ledger balance of the Stack Trading Omnibus corporate accounts</p>                                                                                                                                                                                                                                                                                                               |

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|                                   |      |                             |                                                                                                                                           | <p>across both Forex.com and Interactive Brokers.</p> <p>Check: Returns true only if values match (Anti-Fraud).</p>                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Match Pod Mentor</b>           | POST | user_id, style, asset_class | <pre>`JSON {   "match_found":   bool, "leader_id":   "...", "status":   "Assigned" }</pre>                                                | <p><b>Flow 10 (Pod Onboarding):</b></p> <ul style="list-style-type: none"> <li>• <b>Query:</b> Search Pod_Leader_Registry in SQL.</li> <li>• <b>Criteria:</b> Match Trading_Style + Asset_Class + Status == 'Active' + Current_Count &lt; Max_Capacity.</li> <li>• <b>Outcome A (Match):</b> Increment Leader's Current_Count. Return match_found: true and leader_id.</li> <li>• <b>Outcome B (No Match):</b> Increment Global Pod_Waitlist_Count in SQL. Return match_found: false and status Waitlist.</li> </ul> |
| <b>Execute Level 6 Transition</b> | POST | user_id, country            | <pre>JSON {   "rippling_id":   "[id]",   "contract_url":   "[url]" }</pre>                                                                | <p><b>Flow 6 (W-2/Contractor):</b></p> <p>Lookup: Fetch Base_Income from Table A/B. USA Path: Call Rippling API (/worker.hire) to pre-fill W-2 record using stored KYC data (Name, Address, SSN).</p> <p>Int'l Path: Query database to verify the Dots payout profile is active. Establish the scheduled recurring backend trigger for the new monthly retainer. (No Rippling API action is required, as the International Contractor signed their agreement at Level 1).</p>                                        |
| <b>Calculate Prorated Raise</b>   | POST | user_id, target_level       | <pre>JSON {   "proration_factor"   : float,   "prorated_adjust   ment": float,   "adjusted_retainer   ": float,   "effective_date":</pre> | <p><b>Flow 8 (Pay Raise):</b></p> <p>Data Retrieval: Query Users table for compensation_track and Standard_Hours. Query Matrix Table A/B for Base_Income, Base_Income_Flex, and Base_Income_Type using target_level. Rate Selection: IF</p>                                                                                                                                                                                                                                                                          |

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|                            |      |            | "YYYY-MM-DD" }                                                                               | <p>compensation_track == 'Standard', set Rate = Base_Income. IF compensation_track == 'Flex', set Rate = Base_Income_Flex. Retainer: IF Base_Income_Type == 'Hourly': Adjusted_Retainer = Rate * ((Standard_Hours * 52) / 12). IF Base_Income_Type == 'Annual': Adjusted_Retainer = Rate / 12. Math: Calculate Proration_Factor using Business Days Only (US/Central). Calculate Prorated_Adjustment = Adjusted_Retainer * Proration_Factor. Action: Path A (US W2): Zapier modifies the Rippling compensation rate. Rippling handles native payroll proration on the following cycle. Path B (International): Zapier triggers a one-off Dots API payout for the Prorated_Adjustment AND modifies the active internal recurring payout schedule to the Adjusted_Retainer amount via Dots.</p> |
| <b>Process Pod Vesting</b> | POST | cycle_date | <pre>JSON {   "processed_count": int,   "processed_users": [     array_of_emails   ] }</pre> | <p><b>Flow 17 (Vesting):</b></p> <p>Query: Find all Pod Leaders with Pending_Carry &gt; 0.</p> <p>Routing: Path A (International): Trigger Dots API for one-time payout.</p> <p>Path B (US W2): Trigger Rippling Off-Cycle Payroll API for one-time payout.</p> <p>Immutable Ledger Logging: Within the loop, initialize the Ethers.js wallet and execute contract.recordMetric() natively for each processed leader (metric_category="Pod Leader Vesting", metric_value="Carry Payout Cleared", receipt_hash=[Dots/Rippling Transaction ID]).</p> <p>Cleanup: Set Pending_Carry = 0 to prevent double payment.</p>                                                                                                                                                                           |

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| <b>Process Payment Webhook</b> | POST | provider_signature<br>, payload | HTTP 200 OK | <p>1. Ingest: Accept webhooks from NMI, Triple-A, Nomupay, EBANX, Dusupay, and Skrill.</p> <p>2. Token Capture: IF NMI payload, extract customer_vault_id and save to Users table. Update current_mtd_volume for the specific MID used.</p> <p>3. Parsing: Extract product_id and user_id.</p> <p>4. Routing Execution: IF "RESET": Flatten Positions, Set status='Active_Challenge', Clear trade_history, expiration = Today plus 60, Remove Read-Only. IF "REBUY": Archive current data, Call Provision Sim User logic. IF "EXTENSION": Update expiration_date = expiration_date plus 30. IF "NEW_EVALUATION": Call Provision Sim User logic. IF "MARKET_DATA": Update market_data_paid_until = current_date plus 1 Month.</p> <p>5. External Automation: Push the NEW_PAYMENT_SUCCESS event to Zapier via Flow 1. Zapier parses the metadata to configure market data feeds.</p> <p>6. Broadcast: Broadcast the PAYMENT_RESULT event via WebSocket/Redis to the client. Include user_id, status (SUCCESS/FAIL), and reason_code.</p> <p>7. Audit: Insert record into invoices table. Save product_id into fee_type column.</p> |
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| <b>Generate Daily Snapshots</b><br><br>/financial/daily-snap | POST | none                    | <pre>{ "day": "YYYY-MM-DD", "trader_count": int, "total_pnl": float, "realized_pnl": float, "unrealized_pnl": float, "target_buffer_req": float, "sim_metrics": { "new_sales": int, "resets": int, "pass_rate": float } }</pre> | <b>1. Live Performance Snap:</b> <ul style="list-style-type: none"> <li>• Identify: Query ledgers for all active Live traders.</li> <li>• Group: Categorize Users by level_id.</li> <li>• Sync P&amp;L: For each User, call Rithmic/MT5 API to fetch open_pnl (Unrealized) and closed_pnl_day (Realized today).</li> <li>• Aggregate: Sum Number of Traders per level, unrealized P&amp;L, realized P&amp;L, allocated_capital (Notional), and daily_loss_limit (Stop) per level.</li> <li>• Calculate: target_buffer_req = Level Notional * Global_Var_FCM_Target_Buffer (0.80).</li> <li>• Insert: Create rows in firm_pnl_snapshots.</li> </ul><br><b>2. Sim Funnel Snap:</b> <ul style="list-style-type: none"> <li>• <b>Payments:</b> Count rows in payments table for "Today" where type is 'NEW' vs 'RESET'.</li> <li>• <b>States:</b> Count users where status is 'SIM_ACTIVE'.</li> <li>• <b>Outcomes:</b> Count users where status changed to 'PASSED' or 'FAILED' today.</li> <li>• <b>Duration:</b> Avg(Today's Status_Change_Date minus Sim_Start_Date).</li> <li>• <b>Insert:</b> Create row in sim_funnel_snapshots.</li> </ul> |
| <b>Open for Challenges Status</b><br><br>/system/status      | GET  | Cloudflare Headers Only | <pre>JSON { "allow_signups": boolean, "is_founder_cohort": boolean, "current_price": float, "founder_count": int, "geo_blocked": boolean, "geo_country": "string", "geo_region": "string",</pre>                                | <b>1. Geo-IP Gate:</b> Extract user location from Cloudflare CF-IPCountry and CF-Region headers. Query Compliance_geo_restrictions. If match, set geo_blocked to true and return payload immediately.<br><br><b>2. UI Routing:</b> If not blocked, query checkout_ui_routing. If match, set required_flow; else default to 'FLOW_A'.<br><br><b>3. Gateway Filtering and Localization:</b> Query Payment_Method_Config. Fetch all records where target_country is                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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|                                 |      |                                          | <pre> "required_flow": "string", "methods": [ { "id": "string", "label": "string", "explanatory_text": "string", "cta_text": "string", "icon_tags": ["string"] } }, "is_launch_phase": boolean, "historical_pass_ra te": float } </pre> | <p>'GLOBAL' AND is_active is TRUE. Next, fetch all records where target_country matches geo_country AND is_active is TRUE. Combine these into a single array. Exclusions: IF geo_country is 'IN' for India, remove 'CC' for NMI from the array entirely. The final payload must pass the method_id as 'id', alongside the 'label', 'explanatory_text', 'cta_text', and an 'icon_tags' array mapped from the database for the frontend to render the UI and SVG assets without any hardcoded logic.</p> <p><b>4. Cohort &amp; Pricing Check:</b> Fetch global variables from Zapier Table C for allow_signups and Founder_Cohort_Open. Count rows in Users table where is_founder is true for founder_count. Set current_price based on Founder vs Standard pricing.</p> <p><b>5. Business Logic:</b> Query Platform_Configuration. Set is_launch_phase = TRUE if CurrentDate &lt;= (system_launch_date + 90 days). Fetch cached_pass_rate as historical_pass_rate. Return JSON.</p> |
| <b>Check MID Capacity</b>       | POST | merchant_name, current_mtd_volume        | HTTP 200 OK                                                                                                                                                                                                                             | Internal trigger fired after Process Payment Webhook. Evaluates if current_mtd_volume is greater than or equal to 80 percent of monthly_limit for that specific MID. If true, triggers Zapier Flow 23.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Calculate Volume Pacing</b>  | GET  | None                                     | JSON {<br>"projected_exhaustion": boolean }                                                                                                                                                                                             | Called by Zapier Flow 24. Calculates Average Daily Volume across all active MIDs. Multiplies by total days in month. Returns true if projected volume exceeds total aggregate capacity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Get Notification History</b> | GET  | user_id, status (Optional: Read/Unread), | JSON Array of Objects (notification_id,                                                                                                                                                                                                 | Fetches the paginated history of alerts from the user_notifications table to populate the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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|                               |      | category (Optional),<br>start_date<br>(Optional), end_date<br>(Optional)                                                              | severity, message,<br>timestamp_utc,<br>target_timestamp,<br>category, is_read)                | notification center.                                                                                                                                                                    |
| <b>Mark Notification Read</b> | POST | user_id,<br>notification_id<br>(Optional)                                                                                             | HTTP 200 OK                                                                                    | Updates the is_read flag in the user_notifications table to true. If notification_id is omitted, marks all active notifications as read.                                                |
| Get User Profile              | GET  | user_id                                                                                                                               | JSON Object<br>(profile_photo_url,<br>email,<br>phone_number,<br>ui_preferences)               | Fetches profile and configuration data from the Users table to populate the Account Settings screen.                                                                                    |
| Update User Profile           | POST | user_id,<br>profile_photo_url<br>(Optional), email<br>(Optional),<br>phone_number<br>(Optional),<br>ui_preferences<br>(Optional JSON) | HTTP 200 OK                                                                                    | Updates the provided fields in the Users table. Supports image uploads via S3 signed URL execution before submission.                                                                   |
| Get Market Data Status        | GET  | user_id                                                                                                                               | JSON Array of<br>Objects<br>(feed_name, status,<br>paid_until_date)                            | Evaluates the user's market_data_flags and market_data_paid_until dates to return a structured array mapping the Active, Inactive, or Available status for CME, CBOT, NYMEX, and COMEX. |
| Get Billing History           | GET  | user_id                                                                                                                               | JSON Array of<br>Objects (invoice_id,<br>amount, currency,<br>status, created_at,<br>fee_type) | Fetches paid records from the invoices table.                                                                                                                                           |

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| Get Active Subscriptions                | GET    | user_id                                                                                            | JSON Array of Objects (subscription_id, product_name, next_billing_date, amount)  | Queries recurring NMI subscriptions associated with the user's customer_vault_id for market data add-ons.                                                                                                                                                                                                          |
| Get Saved Cards                         | GET    | user_id                                                                                            | JSON Array of Objects (billing_id, brand, last4, exp_month, exp_year, is_default) | Queries the NMI Customer Vault associated with the user's customer_vault_id to retrieve an array of tokenized credit cards. Returns an empty array if no NMI vault exists.                                                                                                                                         |
| Delete Saved Card                       | DELETE | user_id, billing_id                                                                                | HTTP 200 OK                                                                       | Instructs the NMI Customer Vault to permanently delete the specified tokenized card.                                                                                                                                                                                                                               |
| Capture Lead (Cart Abandonment)         | POST   | email, first_name (Optional), last_name (Optional), utm_source (Optional), abandoned_step (String) | HTTP 200 OK                                                                       | <b>Public Endpoint (No Auth Required).</b><br>Captures partial checkout data triggered by the frontend onBlur event. Validates the email format and saves it to the database as a "Pending Order". Must trigger a webhook to the Zapier automation bus to initiate the Cart Abandonment email sequence.            |
| Provision NinjaTrader Entitlement       | POST   | user_email, status (Active/Inactive)                                                               | JSON { "status": "success" }                                                      | Action:<br>1. API Auth: Authenticate with NinjaTrader Partner API.<br>2. Entitlement: Use the PUT /users/{email}/entitlements method to manually enable or disable the 'Multi-Broker' flag if a user's terminal drops connection.<br>3. Database: Record the NinjaTrader entitlement timestamp in the Users table. |
| Get Market Status /system/market-status | GET    | None                                                                                               | JSON Array of Objects [{ "asset_class": string, "state": string,                  | Query the CME_Instrument_Details table. Calculate the current operational phase for each asset class (Open, Lockout, Closed, Halted) by comparing the server                                                                                                                                                       |

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|  |  |  | "short_text": string,<br>"next_state_timestamp": "ISO8601" ]] | UTC time against the scheduled Official_Close_UTC, Halt_Start_UTC, and Session_Open_UTC. Generate the short_text (e.g. "Close in 45m" or "Market Closed"). Return this array to populate the Market Status Widget on initial dashboard load. |
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**Note (Checkout Integration - Metadata Consistency):** Regardless of the active payment method (NMI, Triple-A, Skrill, Nomupay, Ebanx, Dusupay, Apple Pay, or Google Pay), the Metadata attached to the transaction **MUST** be identical. The downstream Zapier flows (Flow 1, Flow 22) rely on parsing `addon_ids`, `product_id`, and `user_id` from the transaction metadata to correctly provision market data (NYMEX, COMEX, etc). Agency A must ensure all payment gateway payloads are configured to pass these exact custom variables so they can be retrieved seamlessly via the Process Payment Webhook..

## Perks & Batch Process Endpoints

| Endpoint                   | Method | Input Parameters                                               | Output Structure                                                          | Logic Requirement (Precise)                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>Claim Perk (Atomic)</b> | POST   | <code>user_id</code> , <code>perk_level</code><br>(e.g, 1, 10) | JSON { "success":<br>bool, "status":<br>"Claimed" or<br>"Limit_Reached" } | <b>Flow 25 &amp; 26 (Race Condition Protection):</b> <ul style="list-style-type: none"> <li>• <b>Lock:</b> Start DB Transaction.</li> <li>• <b>Check:</b> Query Perks_Registry (Table H). IF Active == Yes AND Claimed &lt; Limit.</li> <li>• <b>Action:</b> Increment Claimed count by 1.</li> <li>• <b>Result:</b> Return true to allow Zapier to trigger the fulfillment (Apliq/Tom James).</li> </ul> |
| <b>Generate Data</b>       | POST   | <code>billing_cycle_date</code>                                | JSON {                                                                    | <b>Flow 22 (Monthly Billing):</b>                                                                                                                                                                                                                                                                                                                                                                         |

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| Invoices                 |      |                      | "invoices_created": int, "errors": [...] }                       | <p><b>Query:</b> Find all Traders where Status == 'Active' AND Asset_Class == 'Futures' AND Market_Data_Billable == True.</p> <p><b>Math (Tiered Billing):</b> Provisional (Level 1 &amp; 2): Invoice_Amount = Total_Market_Data_Cost.<br/>Senior (Level 3+): Invoice_Amount = Total_Market_Data_Cost minus Base_CME_Data_Cost.</p> <p><b>Action:</b> Query database for active MIDs. Retrieve customer_vault_id array. Execute batch NMI charges utilizing Weighted Round-Robin algorithm (weighted by remaining capacity).</p> <p><b>Failure Handling:</b> If charge fails, flag user for "Urgent Payment".</p> |
| Process Defense Recharge | POST | none                 | JSON {<br>"recharged_count": int, "user_ids": [...] }            | <p><b>Flow 21 (Principal Accrual):</b></p> <ul style="list-style-type: none"> <li>• <b>Query:</b> Find Traders where Level &gt;= 18 AND Defense_Used == True.</li> <li>• <b>Math:</b> Calculate Months_Since_Trigger.</li> <li>• <b>Check:</b> IF Months &gt;= 6 AND Status != 'In_Defense':</li> <li>• <b>Action:</b> Set Defense_Used = False (Recharge). Update Last_Recharge_Date.</li> </ul>                                                                                                                                                                                                                 |
| Process Waitlist Queue   | POST | leader_id (Optional) | JSON {<br>"next_candidate_id": "...",<br>"queue_position": int } | <p><b>Flow 11 (Waitlist):</b></p> <ul style="list-style-type: none"> <li>• <b>Trigger:</b> Called when a Pod Spot opens up.</li> <li>• <b>Logic:</b> Query Pod_Waitlist. Sort by Join_Date (FIFO).</li> <li>• <b>Engagement:</b> Return the Oldest Candidate so Zapier can send the</li> </ul>                                                                                                                                                                                                                                                                                                                    |

|                                                   |  |  |  |                                      |
|---------------------------------------------------|--|--|--|--------------------------------------|
|                                                   |  |  |  | "Spot Available" confirmation email. |
| <b>System Status</b><br><br>GET<br>/system/status |  |  |  |                                      |

**Risk Enforcement Endpoints:**

| Endpoint                  | Method | Input Parameters                                                                    | Output Structure                                | Logic Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|--------|-------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Enforce News Event</b> | POST   | asset_class,<br>impact_type (High),<br>stop_threshold (0.70),<br>cap_percent (0.25) | JSON {<br>"scanned": int,<br>"actions": [...] } | <p><b>1. Universal Logic (P&amp;L Threshold - ALL Assets):</b></p> <ul style="list-style-type: none"><li>• Lookup Market_Loss_When_Stopped (Table A/B).</li><li>• IF (Open_PnL) &lt; (-1*(Market_Loss * swap_risk_threshold)):</li></ul> <p><b>Action:</b> FLATTEN_ALL (Close all positions immediately).</p> <p><b>2. Asset-Specific Logic (Size Caps - Only if P&amp;L Safe):</b></p> <ul style="list-style-type: none"><li>• <b>Forex:</b> Lookup Max_Forex_Notional (Table A). Calculate Allowed = Max_Notional * cap_percent.</li></ul> <p>IF Current_Notional &gt; Allowed:</p> <p><b>Action:</b> TRIM_TO_COMPLIANCE passing Target_Limit = Allowed</p> <ul style="list-style-type: none"><li>• <b>Futures:</b> Lookup Max_Contracts (Table B). Calculate Final_Allowed= Floor(Max_Contracts * cap_percent).</li></ul> |

|                                 |      |                                               |                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------|-----------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |      |                                               |                                                          | <p>IF Open_Contracts &gt; Final_Allowed:</p> <p><b>Action:</b> TRIM_TO_COMPLIANCE passing Target_Limit = Final_Allowed</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Enforce Overnight Margin</b> | POST | None (Internal triggered by Node.js Schedule) | <pre>JSON {   "scanned": int,   "actions": [...] }</pre> | <p><b>1. Dynamic Timekeeper and Ingestion Architecture</b></p> <p>The Node.js middleware functions as the timeline authority. It relies on a single daily cron task that extracts schedules from the CME Reference Data API and commits them directly to the <b>CME_Instrument_Details</b> database table. The enforcement engine continuously reads the <b>Official_Close_UTC</b>, <b>Halt_Start_UTC</b>, <b>Halt_End_UTC</b>, and <b>Session_Open_UTC</b> fields for each whitelisted asset family to map out its execution timeline dynamically.</p> <p>Because the CME schedule dictates distinct closing boundaries per asset class, the middleware manages independent timeline threads for each category. For any given asset class, the following sequential trigger milestones are generated daily based on its specific exchange configurations:</p> <ul style="list-style-type: none"> <li>• <b>T-3 Minutes:</b> Asset-Specific Pre Close Entry Lockout Window.</li> <li>• <b>T-2 Minutes:</b> Risk Engine Compliance Audit and Position Trim Execution.</li> <li>• <b>T-30 Seconds:</b> Native Broker Level Failsafe Trigger.</li> <li>• <b>Halt_Start_UTC:</b> Intraday Maintenance Freeze Gateway Command.</li> <li>• <b>Halt_End_UTC:</b> Intraday Maintenance Unfreeze Gateway Command.</li> <li>• <b>Session_Open_UTC:</b> Dynamic Session Opening and Day Margin Restoration.</li> </ul> <p><b>2. Universal Drawdown Safety Gate (Executed at T-2 Minutes)</b></p> <p>At exactly two minutes prior to any asset class close, the Risk Engine evaluates the user's running equity position against their specific tier parameters. This check runs prior to any margin volumetric calculations.</p> <ul style="list-style-type: none"> <li>• <b>The Valuation Formula:</b><br/> <math display="block">\text{Total\_Equity\_Loss} = \text{Account\_Starting\_Equity} -</math> </li> </ul> |

|  |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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|  |  |  |  | <p><b>Current_Account_Equity</b></p> <ul style="list-style-type: none"><li>• <b>The Drawdown Check:</b> The engine checks if the <b>Total_Equity_Loss</b> has reached or crossed 70% of the level's absolute stop loss threshold as written in configuration Table A or Table B.</li><li>• <b>Action:</b> If this 70% loss boundary is breached, overnight holding privileges are revoked entirely. The system dispatches a global <b>FLATTEN_ALL</b> command to the broker execution server, cancelling working orders and clearing all open exposure for that user.</li></ul> <p><b>3. Asset-Specific Pre-Close Lockout Mechanism (Executed at T-3 Minutes)</b></p> <p>At exactly three minutes prior to the specific <b>Official_Close_UTC</b> for a given asset class, the API gateway activates a <b>Pre_Close_Lockout</b> state for those specific symbols.</p> <ul style="list-style-type: none"><li>• <b>Order Filter Rules:</b> The gateway immediately rejects any inbound order payload that would initiate a new position or increase the net directional exposure of an existing position within the closing asset class.</li><li>• <b>Authorized Transactions:</b> The gateway explicitly permits order routing for closing or flattening transactions (such as reducing contract sizes, executing partial fills, or clearing a position entirely).</li></ul> <p><b>4. Futures Asset-Specific Margin Enforcement (Executed at T-2 Minutes)</b></p> <p>If the universal drawdown check passes, the Risk Engine validates the closing asset class against a dual constraint margin calculation. The system utilizes the non-static overnight margin metrics stored via the daily CME cron cycle.</p> <ul style="list-style-type: none"><li>• <b>Required Margin Calculation:</b> The system calculates the capital required to hold the closing contracts overnight. This required value is the greater of either the CME Overnight Margin (Contract Count multiplied by the CME asset margin rate pulled from <b>CME_Instrument_Details</b>) OR</li></ul> |
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|  |  |  |  | <p>25% of the total position notional value of the closing assets.</p> <ul style="list-style-type: none"><li>• <b>Account Equity Validation:</b> The engine verifies if the Available Account Equity (Total Equity minus any margin already encumbered by previously closed asset classes) is greater than or equal to the calculated Required Margin.</li><li>• <b>Action:</b> If the portfolio fails this check, the middleware triggers our internal <b>TRIM_TO_COMPLIANCE</b> algorithm. The engine dispatches automated market close orders directly to the Rithmic gateway, scaling down position sizes step-by-step for the closing asset class until the remaining exposure satisfies the requirements.</li></ul> <p><b>5. Forex Specific Enforcement (Executed at T-2 Minutes to Rollover)</b></p> <p>If the universal drawdown check passes, Spot Forex positions (MT5 and TraderEvolution) undergo an enforcement sweep to lock in compliance before the 16:00 CT (17:00 EST) institutional rollover.</p> <ul style="list-style-type: none"><li>• <b>Exposure Constraint Check:</b> Total open Forex notional exposure must be less than or equal to 25% of the total account equity.</li><li>• <b>MetaTrader 5 (MT5) Core:</b> If a portfolio crosses the 25% threshold, the middleware invokes the server-side YourBourse Contest Plugin API to reduce account leverage limits, allowing the native MT5 Stop Out engine to clear the over-leveraged volume.</li><li>• <b>TraderEvolution Core:</b> The middleware checks for native group leverage reduction functionality. If missing, the Node.js middleware programmatically executes the <b>TRIM_TO_COMPLIANCE</b> algorithm, sending targeted liquidation strings directly to the TraderEvolution Admin API to scale back open lots until the portfolio matches the 25% cap.</li></ul> <p><b>6. Native Broker Failsafe Configuration (Executed at T-30 Seconds)</b></p> |
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|  |  |  |  | <p>To eliminate any execution gap or API latency slippage, the Rithmic core environment must be configured with a hard server-level failsafe.</p> <ul style="list-style-type: none"> <li>• <b>R Admin Configuration:</b> The Rithmic Risk Administrator (R Admin) portal must be configured for each user profile with a "Time-Based Auto-Flatten" parameter set to exactly 30 seconds prior to the specific instrument's exchange halt.</li> <li>• <b>Action:</b> If the Node.js middleware fails to confirm a fill for the required trim, the Rithmic gateway automatically executes a market flatten on the remaining non-compliant positions at the T-30 second mark, ensuring absolute compliance with FCM capital requirements.</li> </ul> <p><b>7. Intraday Halt and New Session Open Operations</b></p> <p>The timekeeper wrapper evaluates halts and re-openings dynamically without hardcoded assumptions:</p> <ul style="list-style-type: none"> <li>• <b>Halt Tracking:</b> At the timestamp written to <code>Halt_Start_UTC</code>, the gateway locks order routing for that symbol and dispatches an outbound payload with <code>Alert_Level = 'Halt_Active'</code>. At <code>Halt_End_UTC</code>, order routing unlocks, and the system dispatches <code>Alert_Level = 'Halt_Cleared'</code>.</li> <li>• <b>Session Opening:</b> At the exact timestamp matching <code>Session_Open_UTC</code> logged in the reference database, the middleware initializes the new trading session for that asset class. <ol style="list-style-type: none"> <li>1. The system drops the heavy CME overnight margin weight calculation and we return to the normal FCM intraday day trading margins.</li> <li>2. The API gateway releases the pre-close lockout filters, returning the account status to standard active execution modes.</li> <li>3. The middleware container fires</li> </ol> </li> </ul> |
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|                             |      |                             |             | <p>an outbound webhook payload to the receiver gateway with <b>Alert_Level = 'Session_Open'</b>, explicitly passing the targeted <b>Asset_Class</b> and affected symbols to refresh user interfaces and broadcast notifications instantly.</p> <p><b>8. Notification Routing Logic</b><br/> At each designated milestone (<b>Warning_60</b>, <b>Warning_30</b>, <b>Warning_10</b>, <b>Lockout_Active</b>, <b>Enforcement_Complete</b>, <b>Halt_Active</b>, <b>Halt_Cleared</b>, <b>Session_Open</b>), the middleware container publishes an outbound data payload to the webhook catcher, transferring control to Zapier Flow 24 to update user dashboard notification interfaces and community broadcast channels instantly. Payload configurations must map the specific symbol strings where applicable to provide accurate tracking clarity to the trading floor.</p> |
| <b>Fail Challenge</b>       | POST | user_id,<br>reason (String) | HTTP 200 OK | <p><b>Atomic Kill Switch (Flow 2):</b></p> <ol style="list-style-type: none"> <li>1. <b>Lock:</b> Disable order entry via Broker API.</li> <li>2. <b>Flatten:</b> Send Market Close for all open positions.</li> <li>3. <b>Cancel:</b> Cancel all working orders.</li> <li>4. <b>DB Update:</b> Set Users.status = 'Failed' and failure_timestamp = NOW().</li> </ol> <p><i>Note: Only returns 200 OK once account is confirmed flat.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Lock &amp; Liquidate</b> | POST | user_id                     | HTTP 200 OK | <p><b>Promotion Safety (Flow 4):</b></p> <ol style="list-style-type: none"> <li>1. <b>Lock:</b> Disable trading temporarily.</li> <li>2. <b>Verify Flat:</b> Check Open_Positions. If &gt; 0, execute Market Close.</li> <li>3. <b>Loop:</b> Retry verification every 500ms (Max 10s) until Open_Positions == 0.</li> <li>4. <b>Unlock:</b> Returns success to Zapier to signal it is</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                     |      |         |             |                                                                                                                                                                                                                                                                                                           |
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|                     |      |         |             | safe to proceed with Funds Transfer.                                                                                                                                                                                                                                                                      |
| <b>Disable User</b> | POST | user_id | HTTP 200 OK | <b>Termination Logic (Flow 7):</b> <ul style="list-style-type: none"> <li>• Call Broker API to disable <b>Execution</b> privileges.</li> <li>• <b>State:</b> Keep Dashboard Access active (User must be able to login to buy Reset).</li> <li>• <b>Do NOT</b> delete the User Record from SQL.</li> </ul> |

## TRIM TO COMPLIANCE

**Definition:** **TRIM\_TO\_COMPLIANCE** Algorithm When the API calculates a Trim is required, it must execute the following asynchronous logic loop server-side, utilizing the **Target\_Limit** passed from the enforcement endpoint. The system must utilize event-driven listeners for execution reports. Hardcoded sleep timers are forbidden.

- Calculate Overage: \* Futures: **Overage = Open\_Contracts - Target\_Limit**
  - Forex: **Overage = Current\_Notional - Target\_Limit**
- Snapshot: Fetch all open positions for the **user\_id**.
- Sort: Order positions by open contract size (Futures) or notional value (Forex) from Largest to Smallest.
- Determine Trim Quantity: Select the single largest open position.
  - Futures: **Order\_Qty = Minimum(Overage, Position\_Contract\_Size)**
  - Forex: **Order\_Qty = Minimum(Overage, Position\_Notional\_Value)**
- Execute: Send **Market\_Close\_Order** for the exact **Order\_Qty** calculated above and capture the **Order\_ID**.
- Event Listener: Monitor the gateway socket for the **ExecutionReport** (Fill Confirmation) associated with the **Order\_ID**:
  - IF Status == "FILLED": Proceed immediately to the Re-Check phase without delay.
  - IF Status == "REJECTED": Instantly halt the loop for this **user\_id**. Push a Severity 1 Alert to the Slack #risk-ops channel containing the gateway rejection reason.
  - FAILSAFE (2000ms Timeout): If the exchange fails to confirm a fill within 2000ms of order submission, abort the trim loop and immediately execute a **FLATTEN\_ALL** command (Hard Liquidation) to protect firm capital.
- Re-Check: Recalculate total **Open\_Contracts** (Futures) or **Current\_Notional** (Forex).
- Loop Condition:
  - Futures: Repeat entire sequence until **Open\_Contracts <= Target\_Limit**.
  - Forex: Repeat entire sequence until **Current\_Notional <= Target\_Limit**.

## Financial & Transaction Endpoints:

| Endpoint                                                   | Method | Input Parameters                                                      | Output Structure                                                                 | Logic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------|--------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Update Account Risk</b>                                 | POST   | user_id, new_level                                                    | HTTP 200 OK                                                                      | <p><b>Flow 4 (Promotion):</b></p> <ul style="list-style-type: none"> <li>• <b>Lookup:</b> Fetch Max_Contracts (Futures) or Max_Notional (Forex) for new_level.</li> <li>• <b>Action:</b> Call Broker API (<b>Rithmic / MT5</b>) to overwrite risk parameters.</li> <li>• <b>Safety:</b> Verify account is Flat via Lock &amp; Liquidate first.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Reset Virtual Ledger</b><br><br>/financial/reset-ledger | POST   | User_id<br>new_level_id (Optional) and deactivate (Boolean, Optional) | { "status": "success",<br>"new_balance": [amount],<br>"pnl_archived": [amount] } | <p><b>Zapier Reference: Flow 6 (Payouts), Flow 7 (Failures)</b></p> <p><b>1. Sync State:</b> Call Execution API (Rithmic, MT5, TraderEvolution) to get final closed P&amp;L.</p> <p><b>2. Update Virtual Ledger:</b></p> <ul style="list-style-type: none"> <li>• <b>Action:</b> Update <code>ledgers.virtual_equity</code> to match the target Reset Balance (e.g. Start Cap).</li> <li>• <b>Log:</b> Insert the difference (Profit/Loss) into <code>historical_pnl</code> table.</li> </ul> <p><b>3. Update Risk Limits:</b></p> <ul style="list-style-type: none"> <li>• <b>Action:</b> Call Execution API to reset <code>Daily_Loss_Limit</code> based on the new balance.</li> <li>• <b>Forex Check:</b> Ensure <code>Max_Notional</code> is re-validated against the level rules.</li> </ul> <p><b>4. Metadata Update</b></p> <ul style="list-style-type: none"> <li>• If new_level_id is provided, update</li> </ul> |

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|----------------------------------------------------------------------|------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                      |      |                            |                                                                                                                                                                                                    | ledgers.level_id. If deactivate is TRUE, set ledgers.is_active = FALSE.                                                                                                                                                                                                                                                                                                                              |
| <b>Process Severance Payout</b>                                      | POST | user_id,<br>reason_code    | JSON {<br>"payout_amount":<br>float, "status":<br>"Sent" }                                                                                                                                         | <b>Flow 9 (Severance):</b><br><br>Logic: Authorized_Amount =<br>Severance_Pay -<br>(Amount_Over_Stop_Loss IF > 100)<br>Safety: Prevents Zapier from<br>performing "Loss Deduction" math on<br>stale data.<br><br>Action:<br>Path A (US Contractor): Triggers Dots<br>API.<br>Path B (International Contractor):<br>Triggers Dots API.<br>Path C (US W2): Triggers Rippling<br>Off-Cycle Payroll API. |
| <b>Calculate Data Refund</b><br><br>/financial/calculate-data-refund | GET  | user_id (UUID)             | {<br>"refundable_amount": float,<br>"currency": "USD",<br>"invoice_count":<br>int }                                                                                                                | Query: SELECT SUM(amount) FROM<br>invoices WHERE user_id = [Input]<br>AND fee_type =<br>'MARKET_DATA_BASE' AND status =<br>'paid'<br><br>Logic: Must calculate the total "Base"<br>data fees paid by the user during<br>their time in Level 1 & 2. Must<br>EXCLUDE any "Add-On" fees (e.g.<br>NYMEX/COMEX) or "Platform" fees<br>which are non-refundable.                                           |
| <b>FCM Health Monitor</b><br><br>/financial/fcm-health               | GET  | none (Scheduled<br>Hourly) | Array of THREE<br>JSON Objects:<br>[<br>{ "institution":<br>"Ironbeam",<br>"fcm_equity":<br>float, "required":<br>float, "action":<br>"DEPOSIT" or<br>"WITHDRAW" or<br>"NONE", "amount":<br>float, | Zapier Reference: Flow 27 (Hourly<br>Solvency Monitor)<br><br>1. Calculate Futures<br>Requirement: Query ledgers<br>for ALL Active Live Futures<br>Traders.<br>Ironbeam_Required_Equity =<br>SUM( allocated_capital ) *<br>Global_Var_Futures_Margin_Ra<br>tio.<br>2. Calculate Forex Requirements<br>(Independent): Because a                                                                       |

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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|  |  |  | <p>"slack_message":<br/>"string", "severity":<br/>"SEV1" or "SEV3" or<br/>"NONE" },</p> <p>{ "institution":<br/>"Forex.com",<br/>"fcm_equity":<br/>float, "required":<br/>float, "action":<br/>"DEPOSIT" or<br/>"WITHDRAW" or<br/>"NONE", "amount":<br/>float,<br/>"slack_message":<br/>"string", "severity":<br/>"SEV1" or "SEV3" or<br/>"NONE" },</p> <p>{ "institution":<br/>"Interactive<br/>Brokers",<br/>"fcm_equity":<br/>float, "required":<br/>float, "action":<br/>"DEPOSIT" or<br/>"WITHDRAW" or<br/>"NONE", "amount":<br/>float,<br/>"slack_message":<br/>"string", "severity":<br/>"SEV1" or "SEV3" or<br/>"NONE" } }</p> | <p>single MT5 account's orders are dynamically routed by lot size, the middleware MUST query the YourBourse Bridge API for the aggregate open notional currently held at each LP. <math>\text{Forex\_Com\_Required} = (\text{Forex.com Open Notional}) * \text{Global\_Var\_Forex\_Margin\_Ratio}</math>. <math>\text{IBKR\_Required} = (\text{IBKR Open Notional}) * \text{Global\_Var\_Forex\_Margin\_Ratio}</math>.</p> <ol style="list-style-type: none"> <li>3. Get Actuals: Call Ironbeam API: Get Net_Liquidating_Value. Call Forex.com API: Get Net_Liquidating_Value. Call Interactive Brokers API: Get Net_Liquidating_Value.</li> <li>4. Evaluate Logic (Executed independently for ALL THREE Institutions): <math>\text{Target\_Line} = \text{Required\_Equity} * \text{Global\_Var\_FCM\_Target\_Buffer} (0.80)</math>. <math>\text{Danger\_Line} = \text{Target\_Line} * \text{Global\_Var\_FCM\_Danger\_Buffer} (0.80)</math>. <math>\text{Excess\_Line} = \text{Target\_Line} * \text{Global\_Var\_FCM\_Withdraw\_Threshold} (1.25)</math>.</li> <li>5. Deposit and Withdrawal Execution (Per Institution):</li> <li>6. Deposit Logic (Chunking): IF <math>\text{Actual} &lt; \text{Danger\_Line}</math>: Set action="DEPOSIT", severity="SEV1". <math>\text{Shortfall} = \text{Target\_Line} - \text{Actual}</math>. <math>\text{Deposit\_Amount} = \text{MAX}(\text{Shortfall}, \\$50,000)</math>. slack_message = "🚨 CRITICAL: [Institution] SOLVENCY RISK. Initiating Emergency Deposit of \$[Deposit_Amount]."</li> <li>7. Withdrawal Logic (Floors): IF <math>\text{Actual} &gt; \text{Excess\_Line}</math> AND <math>\text{Actual} &gt; \\$100,000</math>: <math>\text{Surplus} = \text{Actual} - \text{Target\_Line}</math>. IF <math>\text{Surplus} \geq \\$50,000</math>: Set</li> </ol> |
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|                                                                              |      |                                                                |                                                                 | action="WITHDRAW",<br>severity="SEV3",<br>Withdraw_Amount = Surplus.<br>slack_message ="💰 Surplus:<br>Excess capital detected at<br>[Institution]. Withdrawal of<br>\$[Withdraw_Amount]<br>available."                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Run Capital Rebalance</b><br><br>/financial/run-rebalance                 | POST | force_exec<br>(Bool)                                           | { "status":<br>"executed",<br>"slack_alert_text":<br>[string] } | Zapier Reference: Flow 28 (Hourly Job)<br><br>1. Check Health: Call<br>/financial/fcm-health to retrieve<br>the array containing Futures and<br>Forex objects.<br>2. Execute Logic (Loop through<br>array):<br>• Emergency: IF severity ==<br>"SEV1", Execute Transfer<br>IMMEDIATELY for that Asset<br>Class Institution (FCM or RFED)<br>for the Deposit_Amount.<br>• Scheduled: IF force_exec is<br>True and action ==<br>"WITHDRAW", Execute Transfer<br>for that Asset Class Institution<br>(FCM or RFED) for the<br>Withdraw_Amount.<br>3. Return Notification:<br>• Return an array of<br>slack_messages from both<br>executions to Zapier for internal<br>Ops Channel.<br>• If execution failed, prepend "❌<br>FAILURE:". |
| <b>Execute Treasury Transfer</b><br><br>/financial/execute-treasury-transfer | POST | amount (float),<br>direction (string),<br>institution (string) | { "draft_id":<br>"string",<br>"slack_confirm":<br>"string" }    | 1. Routing: Fetch Mercury Banking<br>API credentials and specific<br>Destination Config based on the<br>specific institution passed in the<br>payload ("Ironbeam", "Forex.com",<br>or "Interactive Brokers").<br>2. Execution: Call Mercury API to<br>generate a DRAFT WIRE transfer<br>(Bank to Institution or Institution<br>to Bank). Do NOT auto-approve.<br>3. Output: Return slack_confirm: "✅<br>Draft Treasury Wire generated via<br>Mercury. Awaiting CEO or CFO                                                                                                                                                                                                                                                         |

|                                                                  |      |                                                                                                                    |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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|                                                                  |      |                                                                                                                    |                                                                                                                                                                                                                        | Approval. Amount: \${amount}, Direction: [direction], Destination: [Institution]. Link: [Mercury_Draft_Link]"                                                                                                                                                                                                                                                                                                                   |
| <b>Generate Promo Code</b>                                       | POST | type,<br>origin_user_id                                                                                            | JSON {<br>"promo_code":<br>"STRING" }                                                                                                                                                                                  | Viral Reward: Generates a single-use, 100% off promo code for an Associate Track or Reset. These are transferable. The system must log the origin_user_id for auditing.                                                                                                                                                                                                                                                         |
| <b>Check Bounty URL</b>                                          | GET  | url (URL Encoded)                                                                                                  | JSON {<br>"is_unique":<br>BOOLEAN }                                                                                                                                                                                    | Anti-Fraud: Queries the bounty_submissions table. Returns false if the URL has already been rewarded to prevent duplicate payouts.                                                                                                                                                                                                                                                                                              |
| <b>Record Immutable Ledger</b><br><br>POST<br>/api/ledger/record | POST | trader_id (string),<br>metric_category (string),<br>metric_value (string), date (string),<br>receipt_hash (string) | JSON { "status":<br>"success",<br>"tx_hash": "string"<br>}                                                                                                                                                             | Validate the Zapier JWT signature.<br><br>Initialize the server-side Ethers.js wallet using the secured environment private key.<br><br>Execute contract.recordMetric(), passing the payload to Polygon via the Alchemy RPC.<br><br>Await confirmation and write the Polygon transaction hash to PostgreSQL.                                                                                                                    |
| <b>Public Ledger Feed</b><br><br>/public/ledger                  | GET  | none                                                                                                               | JSON {<br>"total_verified_milestones": int,<br>"recent_events": [<br>{ "trader_id":<br>"string",<br>"metric_category": "string",<br>"metric_value": "string",<br>"timestamp": "string",<br>"tx_hash": "string"<br>}] } | Fetch: Utilize Ethers.js to query the Polygon smart contract to calculate the total aggregate count of all CorporateMetricVerified event logs ever emitted, AND fetch the array of the 50 most recent logs.<br><br>Cache Layer: To prevent RPC rate-limiting, Node.js MUST cache this combined payload in memory. The cache should only refresh from the blockchain every 5 minutes. The frontend will query this endpoint, not |

|                                                              |      |                                                                                     |                                                                   |                                                                                                                                                                                                                                                                                                                                               |
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|                                                              |      |                                                                                     |                                                                   | the blockchain.                                                                                                                                                                                                                                                                                                                               |
| <b>Record Affiliate Conversion</b><br><br>/impact/conversion | POST | user_id,<br>event_type<br>(e.g. "Level_6_Achievement"),<br>bounty_amount<br>(float) | JSON { "status": "Success",<br>"impact_conversion_id": "string" } | <p>1. Attribution Check: Verify if impact_click_id exists in the Users table for the provided user_id.</p> <p>2. Payout Logic: If attributed, call the Impact.com API to record a new conversion event with the bounty_amount override.</p> <p>3. Audit: Log the Impact conversion ID to the invoices table for financial reconciliation.</p> |
| <b>Register Impact Partner</b>                               | POST | user_id                                                                             | JSON { "status": "Success" }                                      | <p>1. Query the Users table. If is_impact_partner is already TRUE, return Success immediately and exit.</p> <p>2. Impact Integration: If FALSE, call the Impact.com API to register the user as a Partner in the internal Velocity Group.</p> <p>3. State Update: Update the PostgreSQL Users table to set is_impact_partner = TRUE.</p>      |

**Daily Journal and Heatmap Endpoints:**

**Context:** The dashboard requires a "Self-Rating Module" where traders can grade their own discipline and record subjective notes. This data is user-generated and date-based.

| Endpoint            | Method | Input Parameters                                                                              | Output Structure                  | Logic Requirement                                                                                                                                                                              |
|---------------------|--------|-----------------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Create Entry</b> | POST   | date (ISO 8601),<br>rating (Integer 1-5),<br>text_content (String),<br>tags (Array, optional) | HTTP 200 OK<br>(Success) or Error | Validates input and creates a new journal record in the database linked to the user's ID. If an entry already exists for this date, append the text or update the rating (logic should support |

|                          |     |                      |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                            |
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|                          |     |                      |                                                                                                                                                                                                                                                                                                                                                                                   | multiple entries per day or upsert based on business rule).                                                                                                                                                                                                                                |
| Get Monthly Journal Data | GET | user_id, month, year | JSON Object {<br>"aggregate_metrics": { "monthly_pnl": float,<br>"winning_days": int,<br>"losing_days": int,<br>"win_rate": float,<br>"avg_win_day": float,<br>"avg_loss_day": float, "days_traded": int, "avg_rating": string },<br>"daily_records": [ {<br>"date": "YYYY-MM-DD",<br>"daily_pnl": float,<br>"rating_score": int,<br>"stamp_grade": string, "notes": string } ] } | The master endpoint for the Journal tab. The backend must query the historical PnL snapshots to calculate the objective metrics, and merge them with the subjective journal entries from the SQL database. This single payload powers the top metric widgets and the daily calendar cards. |

## Community, Trustpilot and Discord Bot Endpoints

| Endpoint           | Method | Input Parameters | Output Structure                                                                                          | Logic Requirement                                                                                                                                                                         |
|--------------------|--------|------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Get Live Broadcast | GET    | None             | JSON { "youtube_id": string, "is_live": boolean, "title": string, "next_broadcast_timestamp": "ISO8601" } | Fetches the active broadcast configuration from the global variables table. If is_live is false, the frontend utilizes the next_broadcast_timestamp to render the visual countdown timer. |

|                     |      |                          |                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                          |
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| Get Pod Data        | GET  | user_id                  | JSON {<br>"leader_name":<br>"string",<br>"leader_level": int,<br>"discord_channel_url": "URL_string",<br>"join_form_url":<br>"URL_string",<br>"members": [ {<br>"name": "string",<br>"level": int, "status":<br>"string" } ] } | Queries the SQL Database to find the assigned pod_leader_id. If unassigned, populates join_form_url with the ActiveCampaign/Typeform destination. If assigned, returns the leader data, the array of active members, and the direct Discord Channel URL. |
| Create Pod Channel  | POST | user_id,<br>leader_name  | JSON { "channel_id":<br>"string",<br>"channel_url":<br>"URL_string" }                                                                                                                                                          | Instructs the Discord Bot to create a private text and voice channel named pod-[leader_name]. Sets permissions to the assigned pod members. Returns the direct URL.                                                                                      |
| Archive Pod Channel | POST | pod_leader_id            | HTTP 200 OK                                                                                                                                                                                                                    | Instructs the Discord Bot to locate the channel tied to the pod_leader_id, remove all user access permissions, and rename the channel to archived-pod-[leader_name]. Clears the channel_url in the SQL Registry.                                         |
| Send Broadcast Chat | POST | user_id,<br>message_text | HTTP 202 Accepted                                                                                                                                                                                                              | Accepts chat messages from the dashboard UI.<br><br>Validates session auth. Pushes the message_text and user                                                                                                                                             |

|                      |               |                                                            |                              |                                                                                                                                                                                                                                                                                                                                                                                        |
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|                      |               |                                                            |                              | name into the Redis broadcast queue for the Discord Bot worker to consume and batch. Returns 202 immediately to prevent UI blocking.                                                                                                                                                                                                                                                   |
| Revoke All Roles     | POST          | user_id                                                    | HTTP 200 OK                  | Instructs the Discord Bot to use the stored discord_user_id to strip "Pod Member", "Pod Leader", and all Title Roles from the user within the Discord server. Executed during Digital Eviction. If discord_user_id IS NULL in the PostgreSQL Database, the Node.js application must drop the action and return a 200 OK status to Zapier to prevent the automation flow from crashing. |
| Link Discord Account | Slash Command | /link<br>[discord_link_token]                              | JSON { "status": "success" } | Bot native command. Bot reads the token, queries the SQL Database to match the user, writes the discord_user_id to the user record, assigns the "Associate Trader" role, and deletes the token.                                                                                                                                                                                        |
| Manage Discord Role  | POST          | user_id, add_role (String, Optional), remove_role (String, | HTTP 200 OK                  | Instructs the Discord Bot to use the stored discord_user_id to add and/or remove specific                                                                                                                                                                                                                                                                                              |

|                           |      |                   |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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|                           |      | Optional)         |             | roles from the user. If discord_user_id IS NULL in the PostgreSQL Database, the Node.js application must drop the action and return a 200 OK status to Zapier to prevent the automation flow from crashing.                                                                                                                                                                                                                                                                        |
| Process Daily Pod Metrics | POST | pod_id, leader_id | HTTP 200 OK | Internal cron triggered at 11:59 PM. The Node.js bot aggregates messages per pod_id. It fires ONE webhook per pod to Zapier containing: { "pod_id": "string", "leader_id": "string", "total_channel_messages": integer, "total_leader_messages": integer, "raw_chat_log": "string" }. The raw_chat_log must be a single compiled string of the day's chat, chronological, formatted as [Time] [User]: Message, capped at 50,000 characters to prevent LLM context window overflow. |
| Log Calendar Strike       | POST | pod_id, leader_id | HTTP 200 OK | Triggered from Google API Cron. Logs a compliance violation if a Pod Leader fails to invite the notetaker bot. Triggers Discord bot to DM the mentor                                                                                                                                                                                                                                                                                                                               |

|                                         |      |                             |                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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|                                         |      |                             |                                                                                                                                                                           | and writes the weekly Slack report metrics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Analyze Trading Plan                    | POST | user_id,<br>document_s3_url | <b>Immediate Sync:</b><br>HTTP 200 OK<br><br><b>Async Webhook Payload:</b><br>JSON {<br>"mentor_email":<br>"string", "pdf_url":<br>"string", "ai_analysis":<br>"string" } | Triggered by Zapier. Node.js returns 200 OK instantly to prevent timeout. Node.js downloads the PDF/Images from S3 in the background and passes payload to Gemini 1.5 Pro. System Prompt: "Analyze this trading plan. Identify structural strengths and potential risk management weaknesses. Do not dictate strategy; provide supportive, objective feedback a mentor can use to guide the trader. Conclude by explicitly reminding the mentor to encourage the trader to use the daily journaling dashboard." Once complete, Node.js POSTs the Async Webhook Payload to the Zapier Catch URL (TRADING_PLAN_ANALYZED). |
| Generate Trading Activity vs Plan Brief | POST | None (Internal Cron)        | JSON {<br>"mentor_email":<br>"string",<br>"mentee_data": [ {<br>"name": "string",<br>"asset_class":<br>"string", "platform":<br>"string",<br>"weekly_pnl": float,         | Scheduled Cron at 3:00 AM Wed. Queries weekly trade logs, journals, and S3 plan data for all mentees. Passes aggregate payload to Gemini 1.5 Pro to generate ai_brief_text. Compiles                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                                           |      |                                                 |                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                               |
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|                                                           |      |                                                 | "current_balance": float,<br>"distance_to_target": float,<br>"distance_to_stop": float, "ai_brief_text": "string" ] ] } | the AI text alongside the exact quantitative metrics required. Fires ONE webhook to Zapier per Pod Leader containing the array of all their mentees for email distribution.                                                                                                                                                                                   |
| Get Trustpilot Stats<br>/community/trustpilot-stats       | GET  | None                                            | JSON { "score": float, "review_count": int }                                                                            | Proxy endpoint: Middleware calls Trustpilot API using stored BUID. Cache result for 60 minutes to reduce API overhead.                                                                                                                                                                                                                                        |
| Get Testimonials<br>/community/testimonials               | GET  | None                                            | JSON [ { "name": "string", "rating": int, "text": "string", "date": "string" } ]                                        | Proxy endpoint: Fetches latest verified reviews. Backend handles sanitization and date formatting.                                                                                                                                                                                                                                                            |
| Trigger Review Invite<br>/community/trigger-review-invite | POST | user_id, delay_hours (int), admin_id (Optional) | HTTP 200 OK                                                                                                             | 1. Delay Scheduling: Schedule an internal task for NOW + delay_hours.<br><br>2. Suppression Check (At Execution): Upon job trigger, query the Users table. IF trustpilot_reviewed_at < 15 days ago OR trustpilot_invited_at < 30 days ago, HALT.<br><br>3. API Dispatch: Call Trustpilot Invitation API using user_id as referenceId.<br><br>4. State Update: |

|                                                                           |      |                            |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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|                                                                           |      |                            |             | <div>Update<br/>trustpilot_invited_at =<br/>NOW().</div> <div>5. Audit: Log execution<br/>to BPS_Audit_Log.</div>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| TrustPilot Review<br>Received<br>/webhooks/trustpilot<br>/review-received | POST | Trustpilot JSON<br>Payload | HTTP 200 OK | <div>1. Signature Validation:<br/>Verify the payload<br/>signature using your<br/>Trustpilot Webhook<br/>Secret (HMAC).</div> <div>2. Data Extraction:<br/>Parse the payload to<br/>extract referenceld<br/>(maps to user_id) and<br/>created_at<br/>(submission<br/>timestamp).</div> <div>3. State Update:<br/>UPDATE Users table:<br/>Set<br/>trustpilot_reviewed_at<br/>= created_at WHERE<br/>user_id = referenceld.</div> <div>4. Response: Return<br/>200 immediately to<br/>acknowledge receipt<br/>and prevent retries.</div> |

**Risk Metrics Logic Definitions:**

| Metric       | Calculation Logic                                                   | Input Data         | Output Structure |
|--------------|---------------------------------------------------------------------|--------------------|------------------|
| Sharpe Ratio | (Mean Daily Return / Standard Deviation) * sqrt(252). Use a rolling | Daily PnL records. | Float (e.g. 1.5) |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                              |                  |
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|                      | <p>30-day window.</p> <p>Step 1: Fetch the daily net PnL for the last 30 calendar days. Let <math>N = 30</math>.</p> <p>Step 2: Calculate Mean Daily Return. Add all daily PnL values together and divide by <math>N</math>.</p> <p>Step 3: Calculate Standard Deviation. For each day, subtract the Mean from the daily PnL and square the result. Add all these squared numbers together. Divide that sum by <math>(N - 1)</math>. Take the square root of the final number.</p> <p>Step 4: Calculate Sharpe. Divide the Mean Daily Return by the Standard Deviation. Multiply the result by 15.874 (which is the square root of 252 trading days).</p>                                                                  |                              |                  |
| <b>Sortino Ratio</b> | <p><b>(Mean Daily Return / Downside Deviation) * <math>\sqrt{252}</math></b>. Only penalizes negative volatility.</p> <p>Step 1: Fetch the daily net PnL for the last 30 calendar days. Let <math>N = 30</math>.</p> <p>Step 2: Calculate Mean Daily Return. Add all daily PnL values together and divide by <math>N</math>.</p> <p>Step 3: Calculate Downside Deviation. Look only at the days with a negative PnL. Square each negative number. Add those squared numbers together. Divide that sum by the TOTAL number of days (<math>N = 30</math>). Take the square root of that result.</p> <p>Step 4: Calculate Sortino. Divide the Mean Daily Return by the Downside Deviation. Multiply the result by 15.874.</p> | Daily Net PnL snapshots.     | Float (e.g. 1.8) |
| <b>Profit Factor</b> | <p>Step 1: Sum the absolute PnL of all winning trades (Gross Profit).</p> <p>Step 2: Sum the absolute PnL of all</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Current level closed trades. | Float (e.g. 2.1) |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                  |                                                                                                   |
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|                          | <p>losing trades (Gross Loss).</p> <p>Step 3: Divide Gross Profit by Gross Loss. If Gross Loss is 0, return the Gross Profit value to prevent a divide by zero error.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                  |                                                                                                   |
| <b>Win Rate &amp; RR</b> | <p>Win Rate: Count winning trades, divide by total trades, multiply by 100.</p> <p>Risk/Reward: Calculate the absolute Average Win and absolute Average Loss. Divide the Average Win by the Average Loss.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | All closed trades.                               | <p>JSON Object {</p> <p>"win_rate_percent": float,</p> <p>"risk_reward_ratio": float</p> <p>}</p> |
| <b>PnL Breakdown</b>     | <p>Unrealized: Sum of ((Current Market Price minus Average Entry Price) multiplied by Position Size multiplied by Tick Value) for all open positions.</p> <p>Realized: Sum of Net PnL for all closed positions in the current session.</p> <p>Total: Unrealized plus Realized.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Real-time position array + Closed trade history. | <p>JSON Object {</p> <p>"unrealized": float,</p> <p>"realized": float, "total": float }</p>       |
| <b>Pulse Score</b>       | <p><b>0-100 Score.</b> It should reset to 100 (Zen) at the start of every daily trading session (Exchange Reset: 18:00 EST for Futures, 17:00 EST for Forex).</p> <p><b>Deductions:</b></p> <ul style="list-style-type: none"> <li>• <b>-25 Points (Averaging Down):</b><br/>Submitting an order in the same direction on the same symbol while the current Unrealized P&amp;L is negative.</li> <li>• <b>-20 Points (Increasing Risk):</b><br/>Modifying an existing Hard Stop order to move the price further away from the current market price.</li> <li>• <b>-15 Points (Erratic Sizing):</b><br/>Submitting an order where the quantity is &gt; 2x the average quantity of the last 10 fully filled or closed orders (Calculations must ignore</li> </ul> | Execution timestamps and price levels.           | Integer (0-100)                                                                                   |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                            |                                                                                                                                             |
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|                         | <p>partial fills).</p> <p><b>Rewards:</b></p> <ul style="list-style-type: none"> <li>• <b>+5 Points (Pyramiding):</b><br/>Adding to a winning position in the same direction on the same symbol (Unrealized P&amp;L is positive).</li> <li>• <b>+10 Points (Reducing Risk):</b><br/>Modifying a Hard Stop to move it closer to the current market price or locking in profit.</li> </ul>                                                                                   |                                            |                                                                                                                                             |
| <b>Holding Time</b>     | <p>Step 1: Isolate the current level closed trades. Separate them into a Wins array and a Losses array.</p> <p>Step 2: Calculate the average duration in minutes for Wins and Losses.</p> <p>Step 3: To generate the coordinates for the UI bell curves, the Node.js backend must apply a Gaussian Kernel Density Estimation function across the duration arrays. Output 50 evenly spaced X and Y coordinate points from the minimum duration to the maximum duration.</p> | Execution timestamps (Open to Close).      | JSON Object {<br>"avg_win_min": float,<br>"avg_loss_min": float,<br>"wins_distribution":<br>[[x,y]],<br>"losses_distribution":<br>[[x,y]] } |
| <b>Avg Win vs Loss</b>  | <p>Average PnL of all winning trades<br/>Average PnL of all losing trades.</p> <p>Add all winning trade PnL together and divide by the total count of winning trades. Repeat the exact process for losing trades.</p>                                                                                                                                                                                                                                                      | Current level closed trades.               | JSON Object {<br>"avg_win": float,<br>"avg_loss": float }                                                                                   |
| <b>Drawdown Heatmap</b> | Total Account Equity minus the Level's Hard Stop Loss threshold.                                                                                                                                                                                                                                                                                                                                                                                                           | Current Equity, Level Stop Loss parameter. | Float (Currency distance to breach)                                                                                                         |
| <b>Equity Curve</b>     | Time-series data of account balance for charting.                                                                                                                                                                                                                                                                                                                                                                                                                          | Daily Balance snapshots.                   | Array [Timestamp, Value]                                                                                                                    |

|                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                            |                    |
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| <b>Trade Efficiency (MFE Tracking)</b><br><br>Maximum Favorable Excursion (MFE) | <p>Step 1: Node.js tracks a "High-Water Mark" for open positions. The middleware must throttle the evaluation of the incoming Unrealized PnL stream to a maximum of 1 evaluation per position every 2 seconds (2000ms throttle).</p> <p>Step 2: If the throttled Unrealized PnL is greater than the value cached in Redis, overwrite the Redis cache.</p> <p>Step 3: When the trade closes, retrieve this max value. This is the Maximum Favorable Excursion (MFE).</p> <p>Step 4: Calculate Efficiency: Divide the final Realized PnL by the MFE and multiply by 100. If MFE is 0 or negative, Efficiency is 0%.</p> | Unrealized PnL stream from Execution Core. | Float (Percentage) |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------|

## Risk Metrics Endpoints:

| Endpoint          | Method | Input Parameters                | Output Structure                                        | Logic                                                                                          |
|-------------------|--------|---------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Get Equity Curve  | GET    | user_id, timeframe (1D, 1W, 1M) | JSON Array of Arrays [ [timestamp_utc, float_value] ]   | Returns time series data. Triggered on page load and when the user clicks the 1D/1W/1M filter. |
| Get Pulse History | GET    | user_id, limit (Integer)        | JSON Array of Arrays [ [timestamp_utc, integer_score] ] | Returns the historical array of Pulse scores calculated at the end of previous sessions.       |

|                            |     |                                                               |                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                         |
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| Get Core Performance       | GET | user_id                                                       | JSON Object {<br>"gross_profit": float,<br>"gross_loss": float,<br>"win_rate_percent": float, "profit_factor": float, "avg_win": float, "avg_loss": float, "sharpe_ratio": float, "sortino_ratio": float, "holding_time": float, { "avg_win_min": float, "avg_loss_min": float, "risk_mitigation_text": string, "wins_distribution": [[x,y]], "losses_distribution": [[x,y]] } } | Triggered only on page load. Calculates the global metrics for the user's current level. Sharpe and Sortino are locked to a trailing 30 day window. Holding time is locked to the current level history.                                                |
| Get Win Rate Grid Data     | GET | user_id,<br>lookback_count<br>(Integer: 10, 30, 50, 100, 200) | JSON Object {<br>"period_total_trades": integer,<br>"period_win_rate": float,<br>"period_profit_factor": float,<br>"period_avg_win": float,<br>"period_avg_loss": float,<br>"trade_results": [boolean, boolean, ...] }                                                                                                                                                           | Triggered on the Win Rate tab. Returns the recalculated aggregate sub-metrics for the specific lookback window. <b>trade_results</b> is a chronologically ordered array of true/false values used by the frontend to render the red and green dot grid. |
| Get Efficiency Matrix Data | GET | user_id,<br>lookback_count<br>(Integer: 10, 30, 50, 100, 200) | JSON Array of Objects [ {<br>"timestamp_utc": string,<br>"trade_efficiency_percent": float,<br>"risk_reward": float } ]                                                                                                                                                                                                                                                          | Triggered on the Efficiency Matrix tab. Returns a highly optimized payload containing only the exact coordinates required to draw the                                                                                                                   |

|  |  |  |   |                    |
|--|--|--|---|--------------------|
|  |  |  | ] | scatter plot dots. |
|--|--|--|---|--------------------|

## Business Processing Support (BPS) Endpoints:

These endpoints are for internal administrative use. Agency A must secure all endpoints in this section behind an Admin JWT authorization layer. Every POST request must enforce the inclusion of a "reason" and "transcript\_url" in the payload to satisfy the audit protocol.

| Endpoint             | Method | Input Parameters                                            | Output Structure               | Logic Requirement                                                                                                                                                                                             |
|----------------------|--------|-------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Admin Global Search  | GET    | query_string                                                | JSON Array of User Objects     | Indexes across User UUID, Email Address, and Broker Login ID. Returns matching user profiles to hydrate the CRM dashboard.                                                                                    |
| Get System Health    | GET    | None                                                        | JSON Object (Gateway Statuses) | Polls the Rithmic Paper Gateway, Rithmic Live Gateway, MT5 Bridge and TraderEvolution Main Label Server. Returns a Red/Yellow/Green status for each to identify upstream outages before users submit tickets. |
| Reduce Profit Target | POST   | user_id, reduction_amount, admin_id, reason, transcript_url | HTTP 200 OK                    | Recalculates the specific SQL target watermark by subtracting the reduction_amount. Syncs the exact new                                                                                                       |

|                         |      |                                                                  |             |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|------|------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |      |                                                                  |             | integer to the MT5 plugin or Rithmic risk engine immediately. Logs to Audit Table and Slack.                                                                                                                                                                                                                                                                                                                               |
| Inject Goodwill Equity  | POST | user_id, credit_amount, admin_id, reason, transcript_url         | HTTP 200 OK | Adds credit_amount to the virtual_equity ledger. Recalculates the Auto-Liquidate Threshold in SQL and pushes the new limit to the Broker Risk Engine. Logs to Audit Table and Slack.                                                                                                                                                                                                                                       |
| Override Time Extension | POST | user_id, new_expiration_date, admin_id, reason, transcript_url   | HTTP 200 OK | <p><b>System Limiter:</b> The Node.js middleware MUST validate the new_expiration_date. The maximum allowable extension is exactly 14 days from the current expiration date. IF request &gt; 14 days, return HTTP 400 Bad Request.</p> <p><b>Execution:</b> Updates expiration_date in SQL. Automatically whitelists the user ID from the automated Zombie Protocol archiving cron job. Logs to Audit Table and Slack.</p> |
| Admin Pod Reassignment  | POST | user_id, new_pod_id (nullable), admin_id, reason, transcript_url | HTTP 200 OK | <p><b>Cooldown Check:</b> Queries SQL for the user's last pod_reassignment_date. IF the last transfer</p>                                                                                                                                                                                                                                                                                                                  |

|                     |      |                                           |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|------|-------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |      |                                           |             | <p>was &lt; 30 days ago, return HTTP 429 Too Many Requests (Cooldown Active).</p> <p><b>Execution:</b> Updates pod_leader_id in SQL and sets new pod_reassignment_date. Modifies Discord roles via Bot API (removes old, adds new). Fires Zapier webhook to notify Pod Leaders. Logs to Audit Table and standard Slack.</p> <p><b>Security Alert:</b> IF a single Admin ID executes &gt; 3 reassignments in a rolling 24-hour window, the backend must fire a high-priority webhook to a restricted Slack channel: "🔴 High Volume Pod Reassignment Alert [Admin ID]".</p> |
| Send Goodwill Merch | POST | user_id, admin_id, reason, transcript_url | HTTP 200 OK | <p>Inventory Check: Queries Table H Perks_Registry for Level = 'Support'. IF Active == Yes AND Claimed &lt; Limit, proceed. IF Limit Reached, return HTTP 403 Forbidden to the UI with message "Merch Inventory Depleted". Execution: Fires a webhook containing the user's shipping address and</p>                                                                                                                                                                                                                                                                      |

|                       |      |                                           |             |                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------|------|-------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |      |                                           |             | shirt_size to Zapier (Flow 37) to route to Apliiq. Audit: Logs to Audit Table and Slack.                                                                                                                                                                                                                                                                                    |
| Trigger Review Invite | POST | user_id, admin_id, reason, transcript_url | HTTP 200 OK | <p><b>1. Suppression Check:</b> Query Users table. IF <code>trustpilot_invite_d_at</code> &lt; 15 days ago, return 409 Conflict.</p> <p><b>2. API Dispatch:</b> Fire verified invitation request to Trustpilot API using <code>referenceId</code> (user_id).</p> <p><b>3. Audit:</b> Log execution to <code>BPS_Audit_Log</code> (Admin ID, Action, Timestamp, Reason).</p> |
| Admin Hard Reset      | POST | user_id, admin_id, reason, transcript_url | HTTP 200 OK | <p><b>RBAC Gate:</b> Requires 'Level_2' or 'Admin' role in JWT. IF Level_1, return HTTP 403 Forbidden.</p> <p><b>Execution:</b> Bypasses payment gateways. Executes Cancel All Orders and Flatten at the broker gateway. Wipes the SQL PnL ledger. Resets drawdown counters. Reactivates the account. Logs to Audit Table and Slack.</p>                                    |

|                                  |      |                                           |                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------|------|-------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Force Session Kick               | POST | user_id, admin_id, reason, transcript_url | HTTP 200 OK               | <p><b>RBAC Gate:</b> Requires 'Level_2' or 'Admin' role in JWT. IF Level_1, return HTTP 403 Forbidden.</p> <p><b>Execution:</b> Pushes a disconnect command directly to the Execution Gateway (Rithmic/MT5) to force-logout a trapped session. Logs to Audit Table and Slack.</p>                                                                                                                                                                                                                                         |
| Issue Full Refund (Account Kill) | POST | user_id, admin_id, reason, transcript_url | HTTP 200 OK               | <p><b>RBAC Gate:</b> Requires 'Level_2' or 'Admin' role in JWT. IF Level_1, return HTTP 403 Forbidden.</p> <p><b>Execution:</b> Calls existing /issue-refund logic with type="FULL". IF NMI: Executes reverse charge. IF Triple-A: Initiates crypto return flow.</p> <p><b>Termination:</b> Instantly executes the Force Session Kick logic to disconnect the user, and calls Set Account State to 'Terminated' to permanently purge the Rithmic/MT5 sub-account.</p> <p><b>Audit:</b> Logs to Audit Table and Slack.</p> |
| Check Market Data                | GET  | user_id                                   | JSON { "db_entitlements": | Queries the SQL Database for the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                              |      |                      |                                                                                                                      |                                                                                                                                                                                                                                                                                       |
|------------------------------|------|----------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status                       |      |                      | array,<br>"rithmic_status": {<br>"CME": boolean,<br>"CBOT": boolean,<br>"NYMEX": boolean,<br>"COMEX": boolean }<br>} | user's purchased market data entitlements. Pings the Rithmic API to check the active signature status for each exchange. Returns both for UI comparison.                                                                                                                              |
| Resend R Trader Instructions | POST | user_id              | JSON { "status": "success" }                                                                                         | Fires the RESEND_RTRADER_INSTRUCTIONS webhook to Zapier.                                                                                                                                                                                                                              |
| Triple-A Crypto Rescue       | POST | tx_hash, user_id     | JSON { "status": "injected_successfully" }                                                                           | Queries the Triple-A API using the provided tx_hash. IF valid and matching invoice: Construct a synthetic payment success payload and POST locally to the existing /webhook/triple-a endpoint. This ensures standard provisioning flows are triggered without custom database writes. |
| Revoke Pod Leadership        | POST | user_id (Pod Leader) | JSON { "status": "leadership_revoked" }                                                                              | Updates Pod_Leader_Registry status to Permanently_Ineligible. Fires the existing POD_LEADER_TERMINATED webhook to trigger Zapier Flow 15A (Dispersal).                                                                                                                                |

|                             |      |                                                  |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------|------|--------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reset Veriff KYC            | POST | user_id                                          | JSON { "status":<br>"kyc_reset" }                         | Clears<br>Veriff_Failed_Flag and<br>Veriff_Attempts count<br>in the SQL database.<br>Fires the<br>VERIFF_MANUAL_RESET<br>webhook to Zapier.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Escalate to L2              | POST | user_id, ticket_id,<br>reason,<br>transcript_url | JSON { "status":<br>"escalated",<br>"ticket_id": string } | 1. Authentication:<br>Verify Agent JWT<br>(Level 1 or higher).<br><br>2. Zendesk Integration:<br>Call Zendesk API PUT<br>/api/v2/tickets/{ticket_id}.json to add the tag<br>L2_Escalation. IF<br>reason contains<br>"Veriff", "KYC", or<br>"Identity", also add tag<br>Compliance_Review.<br><br>3. Internal Comment:<br>Post a private note to<br>the Zendesk ticket<br>containing the "Reason<br>for Override" and the<br>executing Admin ID.<br><br>4. Audit: Commit the<br>action to the<br>PostgreSQL audit table<br>(Admin ID, User ID,<br>Action Type,<br>Timestamp, Reason,<br>Transcript URL). |
| Terminate Pod<br>Leadership | POST | user_id (Pod Leader)                             | HTTP 200 OK                                               | BPS manual firing<br>endpoint.<br><br>1. Revokes Pod_Leader<br>role from Global UUID.<br>2. Modifies<br>Pod_Leader_Registry<br>status to Terminated.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                         |     |         |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|-----|---------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |     |         |                                                   | <p>3. Zeros out Pending_Carry_Balance (unvested tranches) in DB ledger.</p> <p>4. Writes to BPS_Audit_Log (Admin ID, Action, Timestamp, Reason).</p> <p>5. Fires POD_LEADER_TERMINATED webhook to Zapier to execute mentee dispersal and IT offboarding.</p>                                                                                                                                                                                                                                                                                 |
| Get Available Platforms | GET | user_id | <pre>JSON {   "platforms":     ["string"] }</pre> | <p>Logic: Query the Users table to retrieve the user's current asset_class and platform_selection . If the asset_class is Forex, return an empty array because platform switching is disabled for Forex due to tracking and architecture incompatibility. If the asset_class is Futures, query Table I to fetch all rows where Asset_Class matches Futures and Is_Active is TRUE and Platform_Name does not equal the user's current platform_selection . Return the filtered array of platform strings to populate the frontend support</p> |

|                             |      |                                                            |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|------|------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |      |                                                            |             | dropdown.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Admin Platform Reassignment | POST | user_id, admin_id, target_platform, reason, transcript_url | HTTP 200 OK | <p>RBAC Gate: Requires 'Level_2' or 'Admin' role in JWT. If Level_1, return HTTP 403 Forbidden. Query the Users table to verify the user's asset_class is Futures. If the asset_class is Forex, return HTTP 400 Bad Request because platform changes are disabled for Forex accounts to protect trade history. Execution for Futures: 1. Executes Force Session Kick on the active Rithmic network connection to log the trader out safely. 2. Retains the existing Rithmic credentials and sub_account_id intact to preserve all historical metrics. 3. Modifies the platform_selection field in the Users table to match the target_platform. 4. Fires a webhook to Zapier to dispatch the new platform connection instructions to the trader. 5. Logs the action to the</p> |

|  |  |  |  |                                                                  |
|--|--|--|--|------------------------------------------------------------------|
|  |  |  |  | PostgreSQL audit table and pushes an alert to the Slack channel. |
|--|--|--|--|------------------------------------------------------------------|

### 3.3 Background Workers (Cron Jobs)

- **Task Name:** Pass Rate Aggregator
  - Schedule: Daily @ 00:00 UTC
  - Target Table: Platform\_Configuration
  - Logic Requirement: Execute COUNT(user\_id WHERE status = 'Passed') / COUNT(total\_challenges). Update the cached\_pass\_rate float value in the Platform\_Configuration table (where id = 1). This prevents real-time DB locking during the GET /system/status API calls.
- 
- **Task Name: Mentorship SLA Monitor**
  - Schedule: Hourly
  - Target Table: SLA\_Deadlines
  - Logic Requirement: Execute query for Mentee Trading Plan submissions (72 hours) and Mentor Induction Syncs (5 days). If deadline < 12 hours and unfulfilled, fire webhook to Zapier for warning emails.
  - Task Name: Calendar Compliance Auditor
  - Schedule: Daily @ 01:00 UTC
  - Target API: Google Workspace Calendar API
  - Logic Requirement: Run Domain-Wide Delegation audit on "Pod Leaders" OU. If notetaker@stacktrading.com is missing from a multi-party event, fire POST /api/community/calendar-strike.
  - Task Name: Founding Mentor Defense Bypass
  - Schedule: Continuous
  - Target Table: Users
  - Logic Requirement: Ensure Global\_User\_IDs with the Role\_Flag 'Founding Mentor' are protected via a system variable from automated suspension triggers (e.g., Drawdown Defense) impacting their mentorship status.
- 
- **Task Name: Mentee Organic Survey Monitor**
  - Schedule: Daily @ 02:00 UTC
  - Target Table: Users / Pod\_Leader\_Registry
  - Logic Requirement: Identify mentees who have been in a Pod for > 30 days AND have not received a survey in the last 30 days. Fire the SEND\_MENTEE\_SURVEY webhook to Zapier and update the last\_survey\_date timestamp in the database.

- **Task Name:** Trustpilot Invitation Queue
- **Schedule:** Continuous
- **Target Table:** Users / Scheduled\_Tasks
- **Logic Requirement:** Monitors the internal queue for scheduled **Trigger Review Invite** events. Executes the suppression check *at the moment of dispatch*. This ensures that if a user submits a review while the invitation was "pending" in the queue, the system does not spam them.

- **Task Name: CME Reference Data Ingestion**
- **Schedule:** Daily @ 01:00 UTC
- **Target Table:** CME\_Instrument\_Details
- **Logic Requirement:** Execute a fetch request to the CME Reference Data API (/v3/tradingSchedules). Map the returned Schedule IDs for all symbols currently enabled in the CME\_Whitelist table. Parse the operational hours to define the Session\_Open\_UTC and Official\_Close\_UTC for the current and subsequent session dates. Identify any planned maintenance halts or holiday schedule deviations. This must also retrieve the latest CME overnight margin requirements per instrument. Commit these values to the CME\_Instrument\_Details table to drive the dynamic timekeeper and enforcement routing. **Adam Culver** can provide his code for this specific API connection and data mapping to accelerate the integration.

### 3.4 Required Integrations (Full Scope)

The vendor must build deep integrations for the following services:

| Category      | Service            | Integration Requirement                                                                                                                   |
|---------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Trading       | Rithmic, MT5       | Account creation, risk parameter injection, account locking, and market data provisioning.                                                |
| Connectivity  | BourseYours Bridge | Configuration of routing and aggregation logic for Forex liquidity via MT5 White Label.                                                   |
| Financial GL  | QuickBooks Online  | Automated posting of daily P&L, payouts, and revenue to the General Ledger (Flow 7).                                                      |
| Bonus Payouts | Dots               | Automated mass-payout orchestration utilizing Push-to-Debit, RTP, local fiat rails, and Crypto via a single unified Dots API integration. |

|                           |                                                         |                                                                                                                                                                                                                                                                                          |
|---------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Customer Payments</b>  | <b>NMI, Quaderno, Triple-A, Nomupay, EBANX, Dusupay</b> | NMI Gateway integration utilizing the Customer Vault API for tokenization. Quaderno API for real-time global tax calculation. Triple-A for crypto processing. Nomupay, EBANX, and Dusupay integrations utilizing hosted desktop pop up overlay interfaces for localized dynamic routing. |
| <b>HR &amp; Payroll</b>   | <b>Rippling</b>                                         | "Zero-Entry Onboarding" to pre-fill W-2 employee records using captured KYC data.                                                                                                                                                                                                        |
| <b>Identity</b>           | <b>Jumio / Veriff</b>                                   | Automated identity verification gating the transition from Sim to Live.                                                                                                                                                                                                                  |
| <b>Risk Data</b>          | <b>Economic Calendar API</b>                            | A REST endpoint to ingest High-Impact News webhooks (e.g. FXStreet) and trigger risk checks (Flow 23).                                                                                                                                                                                   |
| <b>Affiliate Tracking</b> | <b>Impact.com</b>                                       | Integration via Zapier and API to sync NMI and Triple-A conversion events. Manages both the external enterprise affiliate marketplace and the internal Velocity/Amplify reward tiers via separate contract groups.                                                                       |
| <b>Public Ledger</b>      | <b>Polygon PoS &amp; Alchemy API</b>                    | Deployment of a state-less smart contract and REST API middleware to write immutable logs via Web3 webhooks.                                                                                                                                                                             |

### 3.5 The Automation Bus (Zapier)

**Zapier** will function as the orchestration engine.

- **Configuration:** The environment must be configured at the Company Tier.
- **Logic Matrix:** Zapier Tables will serve as the configuration matrix (Tables A-F in Zapier Doc), allowing operations staff to adjust global variables without code deployment.

### 3.6 Immutable Ledger (Proof of Payout)

Stack Trading requires the implementation of a one-way "Proof of Payout" cryptographic ledger. This system functions as a decentralized receipt printer to publicly verify that trader payouts and corporate milestones have been successfully processed. Vendors are explicitly forbidden from scoping custom blockchain networks, dedicated node infrastructure, or decentralized finance (DeFi) transaction routing. Fiat currency movement remains entirely off-chain.

#### 3.6.1 Smart Contract Architecture (Future-Proofed)

Agency A will develop and deploy a single Solidity smart contract to the Polygon PoS Mainnet.

To eliminate state storage gas costs, the contract must **not** use state variables to store data. All data must be written exclusively to the blockchain transaction logs using a generalized EVM event. The exact required Solidity implementation is as follows:

```
// SPDX-License-Identifier: MIT

pragma solidity ^0.8.0;

contract StackTradingLedger {

    address public owner;

    // Generalized event to capture ANY corporate metric permanently

    event CorporateMetricVerified(string traderId, string metricCategory, string metricValue, string
    timestamp, string verificationHash);

    constructor() {

        owner = msg.sender;

    }

    function recordMetric(string memory _traderId, string memory _metricCategory, string
    memory _metricValue, string memory _timestamp, string memory _verificationHash) public {

        require(msg.sender == owner, "Unauthorized to write to ledger");

        emit CorporateMetricVerified(_traderId, _metricCategory, _metricValue, _timestamp,
        _verificationHash);

    }

}
```

### 3.7 Payment, Tax, and Billing Architecture

The system requires a highly resilient, multi-gateway payment architecture designed for high-risk processing.

#### Core Systems

- **Gateway - NMI:** The Node.js middleware must interface with the NMI Gateway API, which acts as the routing hub for the underlying merchant accounts.
- **Tax - Quaderno:** The middleware must utilize the Quaderno API to monitor global tax thresholds (EU VAT, APAC GST). It must calculate and apply tax at checkout *only* when economic nexus is triggered.
- **Gateway - Triple-A:** The system must support Triple-A for international clients or overflow volume. This is for customers paying via Crypto (we only receive Fiat)
- **Numupay, Dusupay, Ebanx**

## Dynamic MID Load Balancing and Observability

- **Velocity Load Balancing:** The middleware must query the database to distribute transactions across all active Merchant IDs (MIDs) utilizing a Weighted Round-Robin algorithm based on their remaining capacity (monthly limit minus current month-to-date volume). This keeps all accounts consistently warm and prevents risk-flagging from sudden volume spikes.
- **Alerting (80 Percent Warning):** When any single MID reaches 80 percent of its database-defined monthly cap, the backend must fire a webhook to trigger a standard Slack alert to the Operations channel.
- **Alerting (Severity 1 Aggregate Exhaustion):** A daily backend cron job must calculate the Average Daily Volume across all active MIDs. If the Projected End of Month Volume (Average Daily Volume multiplied by Days in Month) is greater than or equal to the Total Aggregate Capacity defined in the database, the system must fire a Severity 1 Critical webhook to trigger a Slack alert.

## 3.8 MT5 Plugin/Server-side code required

MT5 Automations does not appear to support this natively.

### Functionality Requirements:

**IF** A) Hard Stop Loss Breach (realized + unrealized pnl, **or** open + closed equity level)

**OR** B) Target Profit Reached (realized + unrealized pnl, **or** open + closed equity level)

### **THEN (Execution Routine)**

Step 1: Freeze the subaccount (shift to Read-Only to block new order entry).

Step 2: Cancel all open/pending orders.

Step 3: Liquidate all open positions at market.

Step 4: Remain in a read-only state until we send a command to MT5 API to either re-enable with new parameters (move groups) or archive the account.

### First few levels of Progression Configuration:

| Group Parameters  | Notional | Stop Equity Trigger | Stop PnL | Target PnL | Target Equity Trigger |
|-------------------|----------|---------------------|----------|------------|-----------------------|
| Evaluation / Demo | \$25,000 | \$23,125            | \$1,875  | \$3,500    | \$28,500              |
| Provisional (L1)  | \$25,000 | \$23,750            | \$1,250  | \$1,875    | \$26,875              |
| Provisional (L2)  | \$50,000 | \$47,500            | \$2,500  | \$3,750    | \$53,750              |

For example, if a subaccount in L1 reaches Equity of \$23,125 OR \$28,500 we want to trigger the routine.

## 4.0 Technical Constraints & Assumptions

### 4.1 Risk Engine Constraints ("Trim to Compliance")

The system must enforce a "Soft Edge" liquidation policy with the following safeguards (Note: Trim to Compliance applies to Futures/Rithmic ONLY. Forex liquidation is handled natively by Your Bourse):

- Event Driven: The logic must utilize an event listener for the ExecutionReport (Fill Confirmation) to trigger immediate re-calculation of equity.
- Latency Awareness: The calculation logic (hosted in AWS) must account for the ~20ms round-trip latency to the Chicago Exchange.
- Node.js Fail-Safe: The system must implement a 2,000ms timeout. If the exchange fails to confirm a "Trim" fill within this window, the system must default to a "Hard Liquidation" (Flatten All) to protect firm capital.
- Execution Failsafe: Rithmic R|Admin must be configured with a hard temporal trigger 30 seconds prior to market close to ruthlessly Flatten All if the Node.js trim fails or if the sub-account balance drops below CME maintenance margin.

- **Independent:** This needs to be triggered capable of being non blocking and potentially run with concurrency. This may need to be set to its own EKS pod for concurrency or a lambda/step function

## 4.2 Compute & Logic Constraints

- **Node.js Workers (Stateless):** All complex business logic (e.g. Challenge Passing, Drawdown Calcs) must be executed in Stateless Node.js Workers (AWS EKS Fargate).
- **No Stored Procedures:** The use of PostgreSQL Stored Procedures for business logic is prohibited. RDS must remain a passive state store.

## 4.3 Audit & Compliance Constraints

- **Virtual Ledger (Master/Sub-Account Support):** Since the firm operates Proprietary Master Accounts at the FCMs for Futures (Ironbeam) and Stack Trading Omnibus corporate accounts for Forex (Ripple Prime), the internal database maps to individual Sub-Account IDs provisioned on the execution servers.
- **Immutable Archival (S3 Object Lock):** To satisfy regulatory audit requirements, the system must archive the following granular data points to Amazon S3 with Object Lock (WORM compliance) enabled:
  - **Order History:** Full lifecycle events (Placement, Modification, Cancellation) including Timestamp (UTC), Unique Order ID, Limit Price, and Quantity.
  - **Execution Log:** All fill confirmations including Execution Price, Quantity, and Exchange Order ID.
  - **Risk Triggers:** A specific log entry for any system-generated intervention (e.g. *"Daily Loss Limit Breached -> System Triggered Flatten Command"*).
  - **Support Action Audit Trail:** All administrative overrides executed via a Support Screen must be written to an immutable PostgreSQL audit table. The record must include the Admin ID, User ID, Action Type, Timestamp, Reason, and Transcript URL.
  - **Support storage levels:** Must support configuration for differing storage buckets of warm and cold storage.
  - **Retention:** CFTC regulation 1.31 requires 5 year record retention. However only 2 years have to be immediately accessible for inspection. Moving objects > 25 months to the cold storage bucket for long term storage.
- **NFA Audit Requirements:** CFTC Regulation 1.31 ensure audit trails for member activities must be maintained
  - **Scope:** all formal communication, transaction records, promotional materials
  - **Retention:** 5 years and 1 month

## 4.4 Additional Constraints:

4. **Cloud (Futures):** Agency A is fully responsible for provisioning and configuring the AWS

Cloud Environment (EKS/Fargate, RDS, VPC) via Terraform.

5. **Cloud (Futures):** Agency A will configure deployment to redeploy or upgrade with 0 downtime.
6. **Bare Metal (Forex):** The Client (Stack Trading) will lease the Dedicated Windows Server in NY4 directly to maintain the asset. Agency A is responsible for the technical configuration, software installation (MT5 White Label/BourseYours), and establishing the AWS Direct Connect private fiber link to the AWS environment.
7. **Handover:** Upon project completion, Agency A must transfer to the Client: (1) Terraform/IaC repositories and state handover procedure, (2) documented deployment runbooks, (3) all non-human service credentials via the approved secrets manager, and (4) administrator access via IAM role delegation. AWS Root credentials must remain under Client control at all times; the vendor must not request or retain root access. A “break-glass” access procedure must be documented.
8. **API Sandboxes:** The Client (Stack Trading) will provide valid Sandbox/UAT credentials for StoneX, Rithmic, and MT5. The Agency is not expected to purchase these licenses.
9. **Accounting Map:** The Client will provide the specific General Ledger Mapping (Chart of Accounts) for the QuickBooks integration.
10. **Operating System:** Vendors must specify the required OS for the API Wrappers (Windows Server vs. Linux) in their proposal. Agency A is responsible for OS management during the build phase.
11. **API Definition:** Agency A is responsible for defining the OpenAPI (Swagger) Specification for the Middleware Hub and providing it to Agency B (Frontend) for approval prior to the build.
12. **Audio Security:** The Squawk Relay stream must use token-based authentication to prevent unauthorized listening.
13. **DNS Proxy Separation:** Edge Routing / Proxying Strategy: The frontend dashboard domains must be delivered via a global CDN/edge security layer. For the API domain (api.stacktrading.com), the vendor must propose and implement a consistent edge strategy that supports **global performance, WAF/DDoS protection, and webhook reliability**.
14. Acceptable architectures include:
15. A) **Cloudflare as reverse proxy/WAF** for api.stacktrading.com (Orange Cloud), optionally using Global Accelerator as the origin target (Cloudflare → Global Accelerator → ALB).
16. B) **AWS-native edge** for api.stacktrading.com (e.g., Global Accelerator/CloudFront + AWS WAF/Shield), with Cloudflare used for frontend only, and all Cloudflare-header dependencies replaced with AWS-native or server-side GeoIP logic.
17. The RFQ functional requirements (geo gating, webhook allowlisting, rate limiting) must remain satisfied regardless of the chosen design.
18. **Webhook Whitelisting:** Agency A must configure Cloudflare WAF rules to explicitly whitelist the inbound IP ranges for all external providers (Payment Gateways, Veriff, Dots, and the Economic Calendar API). Default WAF settings will falsely identify high-velocity incoming webhooks as DDoS attacks and drop the payloads, causing critical state

failures in the database.

19. **Webhook Blacklisting:** Webhook & Bot Traffic Controls: The edge/WAF layer must explicitly block known Tor exit nodes and high-risk anonymization networks for *public/auth endpoints*. For webhook receiver endpoints, implement signed webhook verification (HMAC/JWS).
20. **Layer 7 Rate Limiting:** Rate limiting must be applied at the Cloudflare edge for all authentication endpoints, form submissions, and webhook catchers to prevent malicious bot traffic from exhausting Zapier task quotas or AWS compute resources.
21. **The Desk Manager Delivery:** Because LLM inference introduces 1 to 2 seconds of latency, the commentary string must be delivered asynchronously via the GEKKO\_CHATTER channel. Under no circumstances should the backend wait for the LLM response before confirming a trade execution or updating the account balance.

#### 4.4 Additional Constraints:

1. **High Availability (AWS):** All production services must be designed for multi-AZ resiliency (EKS/ECS across  $\geq 2$  AZs, RDS Multi-AZ, stateless services behind ALB/Ingress).
2. **DR Targets (Admin Control Plane):** Vendor must propose an AWS DR approach with explicit **RTO/RPO** targets and a documented restore procedure (tested). At minimum, implement automated backups + point-in-time restore for Postgres and versioned/immutable storage for audit logs.
3. **Connectivity Resilience (NY4):** Vendor must document the failover design for AWS  $\leftrightarrow$  NY4 connectivity. If Direct Connect is used for production, include a backup path (e.g., site-to-site VPN) and operational failover procedure.
4. **Operational Tooling:** Must include centralized logging, metrics dashboards, and alerting (CloudWatch dashboards/alarms are acceptable) with documented runbooks for Sev-1 incidents.
5. **Cost Controls:** Vendor must implement cost allocation tags, budgets/alerts, environment separation (dev/stage/prod), and autoscaling policies. Vendor must propose a cost-optimized architecture option (e.g., ECS vs EKS, Cloudflare vs GA/CloudFront) while meeting latency/security goals.
6. **Security Controls:** Vendor must implement security at all levels during the lifecycle of the contract. Authentication & Access security for all deployed assets and underlying technologies, Network Security and hardening to make sure the public does not have access or routes to sensitive systems or interfaces while still allowing stack trading and contractors access to sensitive systems and interfaces. Layer 7 Security for protections of all underlying assets with layer 7 access. Security scanning on all code before deployment/mergers. Threat detection implemented across all technologies so as anomalous and abusive behavior is promptly reported for incident response. Application security logging where all CRUD actions are recorded.

## 5.0 Execution Schedule

| Phase                | Scope                | Key Deliverables                                                                                                                           |
|----------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 1: Foundation  | Infrastructure Setup | AWS setup (RDS, S3, Global Accelerator). Zapier Company Tier config. Connectivity with StoneX.                                             |
| Phase 2: Core Logic  | Logic Implementation | Delivery of API Stubs (Swagger Spec). Implementation of Sim Engine, BourseYours Configuration, Live Onboarding Logic, and Financial Flows. |
| Phase 3: Integration | Testing              | End-to-End testing: User Buy -> Sim Account -> Pass Challenge -> Live W-2 Contract Generation -> Trade Execution. All flows/scenarios.     |
| Phase 4: Launch      | Deployment           | Load testing (500 concurrent users). Soft Launch to Founding Cohort.                                                                       |

## 6.0 Vendor Qualifications

We invite quotations from firms demonstrating specific expertise in:

1. **Capital Markets Architecture:** Experience with OMS/EMS logic and L1/L2 Market Data structures.
2. **API Integration:** Specific track record integrating Rithmic, MetaTrader 5 (MT5) and TraderEvolution.
3. **Backend Engineering:** Deep expertise in Node.js middleware development and PostgreSQL database design.
4. **Infrastructure Orchestration:** Demonstrated ability to manage hybrid environments, specifically routing AWS-hosted Node.js applications to external execution gateways (Rithmic via AWS backbone) and provisioning AWS Direct Connect private fiber circuits to Dedicated Bare Metal Windows Servers in Equinix NY4 (Forex)

## 7.0 Submission Requirements

We request a proposal containing:

1. **Fixed Price Bid:** Total cost to build and deploy the complete scope
2. **Monthly Retainer Quote:** Cost for ongoing managed services and Level 3 support post-launch (Year 1).
  - **SLA Requirement (Tiered):**
    - **Severity 1 (System Down/Failure):** 30-Minute Response Time (24/5 Market Hours).
    - **Severity 2 (General Support):** 4-Hour Response Time.

- **Scope:** The retainer covers maintenance, uptime monitoring, and minor changes/logic adjustments. Major feature development will be scoped separately.
- 2. **AWS Run-Rate Estimate:** Provide an estimated monthly AWS cost model for dev/stage/prod, including major services (compute, DB, storage, data transfer, monitoring) and the assumptions used. This model should support entering usage estimates, and provide estimates based on those entry values.